

The Distorting Effects of Rules of Origin: A Firm-Level Analysis*

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June 2026

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Abstract

Recent decades have witnessed the proliferation of free trade agreements (FTAs), which make free trade between members conditional on compliance with rules of origin (RoO). We develop a model to study firms' decisions to comply with RoO. We test the model's predictions using Brexit as an exogenous shock that led to the introduction of RoO in EU-UK trade relations. We construct a unique dataset on the RoO contained in the EU-UK Trade and Cooperation Agreement (TCA), which we combine with rich administrative data from France. Exploiting within-firm cross-product and cross-country variation in treatment over time, we find that TCA RoO distorted sourcing decisions of French firms, reducing the probability that they source inputs from non-TCA countries. By distorting sourcing decisions, RoO decrease firms' sales, employment, and productivity. Combining the structure of model with our estimates implies a 9.7% increase in expected input sourcing costs for firms complying with TCA RoO.

JEL classification: F14, F23.

Keywords: Rules of origin, TCA, heterogeneous firms, input sourcing.

*We thank Etienne Guigue, Isabelle Mejean, Mathieu Parenti, Mattia Nardotto, and participants at the 2025 ETSG conference and seminar participants at the University of Sydney School of Economics, Erasmus School of Economics, University of Bologna, University of Kyoto, the CESifo Global Economy conference, and the CEPR ERWIT conference for helpful comments. Conconi gratefully acknowledges financial support from the European Research Council (Advanced ERC Grant Agreement No. 834253). Orefice gratefully acknowledges financial support from French National Agency for Research, project GLOWORN.

1 Introduction

Recent decades have witnessed two key trends in international trade. The first is the emergence of global value chains: firms have fragmented their production processes across countries, sourcing many inputs from suppliers around the world (e.g., [Johnson and Noguera, 2012](#); [Antràs and Chor, 2021](#)). The ability to use foreign inputs has substantial effects on firms' productivity (e.g., [Halpern et al., 2015](#); [Blaum et al., 2018](#)). The second trend is the proliferation of regional trade agreements: as of the end of May 2026, 383 such agreements were in force, most of which take the form of free trade agreements (FTAs).¹

FTAs can distort firms' sourcing decisions through preferential tariffs: inputs imported from FTA partners face lower tariffs than inputs imported from third countries.² This channel has been well documented in studies of the trade and welfare effects of regional agreements. For example, [Caliendo and Parro \(2015\)](#) examine the impact of preferential tariff changes among members of the North American Free Trade Agreement (NAFTA). FTAs can also distort firms' sourcing strategies through a second key channel: preferential rules of origin (RoO). These rules are used to distinguish goods originating from member countries from those originating from third countries. In principle, RoO are meant to prevent trade deflection, i.e., to ensure that goods being exported at preferential rates from one FTA partner to another truly originate from the FTA area and are not simply assembled from components originating from third countries. In practice, they also prevent final good producers from choosing more efficient input suppliers outside of the FTA area, in order to avoid losing origin status and the tariff preference it confers. These regulations can give rise to trade diversion in intermediate goods (e.g., [Grossman, 1981](#); [Conconi et al., 2018](#); [Head et al., 2024](#)). In a study by the [ITC \(2015\)](#), RoO emerge as the most problematic non-tariff measure faced by manufacturing firms. This is the first paper to show that preferential RoO distort firms' sourcing decisions, leading them to switch away from suppliers outside the FTA, with negative effects on productivity and exports.

Defining the origin of goods is difficult in a world in which production processes are fragmented across countries. RoO in FTAs use changes in tariff classification (CTC) requirements, which restrict the sourcing of particular inputs (within a chapter, heading, or subheading of the Harmonized System product classification).³ CTC are often combined

¹These correspond to 632 notifications from WTO members, counting goods, services and accessions separately (WTO Secretariat).

²Regional trade agreements constitute an exception to Most Favored Nation (MFN) principle stipulated by Article I of the GATT, according to which a country should grant equal treatment to all imported goods, irrespective of their origin. Preferential agreements are allowed under Article XXIV of the GATT (or under the Enabling Clause for trade agreements involving developing countries).

³For example, the RoO contained in the NAFTA agreement stipulate that watches (heading 91.02 in the

with value added (VA) requirements, which limit how much foreign content (non-originating materials) can be used to qualify for preferential tariffs.

To study firms’ decisions to comply with RoO and the effects of these sourcing restrictions, we develop a model of trade between three countries (Home, Foreign, and Rest of the World). In this model, heterogeneous firms produce differentiated varieties of a final good using labor and a composite intermediate input. Firms located in the Home country face a common distribution of the cost of sourcing the intermediate input from a country. After deciding whether to pay a sunk entry cost to setup production, each firm draws its intermediate input cost. Depending on the observed draw, it chooses from which country to optimally source the intermediate input bundle.

A FTA between Home and Foreign implies that trade between these countries is duty free, conditional on complying with the RoO of the agreement. Firms’ compliance gains depend on the benefits of obtaining duty-free access to the FTA partner’s market for their final good. Firms’ compliance costs are associated with the increase in production costs resulting from distorting input sourcing. In our model, firms for which the compliance gains outweigh the compliance costs stop sourcing the intermediate input from the Rest of the World. A second set of predictions concerns firms’ revenues. Our model suggests that, after the entry into force of the FTA, firms that have to adjust their input sourcing decisions to comply with RoO (i.e. “constrained compliers”) experience an increase in their marginal cost and hence a fall in global revenues.⁴

To assess the validity of these predictions, we leverage Brexit as a shock that led to the introduction of RoO in trade relations between the European Union (EU) and the United Kingdom (UK). Brexit was an unprecedented trade shock: “No government has ever done this kind of reverse co-operation trade agreement before applying the brakes to the EU’s integrationary juggernaut overnight“ (*Financial Times*, January 8, 2021). Rather than remaining in the EU Customs Union (CU), the UK has negotiated an FTA with the EU, the Trade and Cooperation Agreement (TCA). As a result, since January 2021, free trade between the UK and the EU is conditional on compliance with TCA RoO. Producers in the EU (UK) have two options: complying with TCA RoO, in which case they can export their final goods duty free to the UK (EU), but have to stop sourcing restricted inputs from non-TCA countries; or not complying with TCA RoO, in which case they can source restricted inputs

HS classification) must undergo change of HS chapter, i.e. non-originating inputs must not fall under HS chapter 91. This rule implies that watches can only be traded duty free among NAFTA members if the watch movements (HS 91.08), watch straps (HS 91.13) and watch cases (HS 91.12) used to produce them are sourced from producers located within the FTA.

⁴The increase in the marginal cost due to a sub-optimal sourcing strategy leads to higher price and hence to a reduction in total sales.

from non-TCA countries, but face MFN tariffs when exporting their final goods to the UK (EU).

We construct a unique dataset that allows us to map the input-output linkages in the RoO embedded in the text of the TCA. To this purpose, we follow the methodology used by [Conconi et al. \(2018\)](#) to code the RoO contained in the NAFTA agreement. For every final good, we trace all the inputs that are subject to RoO requirements in the TCA. Similarly, we link every intermediate good to the final goods that impose RoO restrictions on its sourcing.

We merge this novel dataset with rich administrative firm-level data (annual accounts and customs records) for France. This allows us to trace the inputs a French firm cannot source from outside the TCA (e.g., the United States, China, Japan, Vietnam), if it wants to continue exporting duty free to the UK after January 2021. We use tariff data from Market Access Map (MAcMap-CEPII) to measure the tariffs faced by French firms when importing inputs from different countries. We also measure a firm’s compliance gains, which depend on the preferential margin — captured by the tariff the firm faces when exporting its final goods to the UK if it does not comply with TCA RoO — and on the importance of the UK as an export market — captured by the firm’s share of exports to the UK pre TCA.

We first assess the validity of our model’s predictions concerning the effects of RoO on firms’ sourcing decisions. One may be concerned about the endogeneity of the sourcing restrictions included in the TCA. Interestingly, the RoO contained in the TCA are mostly inherited from previous EU FTAs, which attenuates endogeneity concerns.⁵ Our identification strategy relies on a triple-difference methodology, exploiting within-firm cross-product and cross-country variation in RoO treatment over time. This approach allows us to include a battery of fixed effects (firm-country-year, firm-product-year and product-country-year in the most demanding specifications) to address concerns about omitted variable bias. We also carry out placebo tests, verifying that (i) RoO only affect sourcing decisions if firms have incentives to comply with them, and (ii) the entry into force of the TCA affects sourcing from non-TCA countries only through RoO.

The analysis demonstrates that the introduction of TCA RoO has significantly reduced the probability that French firms source inputs from non-TCA countries. The results are robust to different fixed effects and econometric methodologies. Our estimates imply that a one standard deviation increase in the number of inputs restricted by TCA RoO decreases the average probability of sourcing from non-TCA countries by up to 29%.

We also estimate event-study specifications, which allow us to examine possible anticipa-

⁵For example, many of the rules contained in the TCA coincide with those in the FTA between South Korea and the EU (KOREU), which have been coded by Gallina *et al.*, 2025. For rules on goods ranging between HS 01.01 and HS 01.06, the RoO in KOREU reads “ All the animals of Chapter 1 shall be wholly obtained”; the corresponding rule in the TCA is “All animals of Chapter 1 are wholly obtained.”

tion effects and variation in treatment effects since the entry into force of the TCA in 2021. We find that the negative effects of RoO become larger over time, suggesting that it takes time for firms to change input suppliers. We find no significant pre-trends, indicating that firms did not adjust their sourcing decisions in anticipation of RoO before the entry into force of the TCA.

The results of the placebo tests confirm that RoO distort firms’ sourcing decisions only when firms have incentives to comply with these rules. As expected, we find that the trade diverting effects of RoO on imports of intermediate inputs are larger for firms with larger compliance gains. We also provide evidence of heterogeneous effects across inputs: firms are more likely to stop importing inputs from non-TCA suppliers when these are more substitutable across countries, as measured by [Fontagné et al. \(2022\)](#).

We next examine the effects of RoO on firms’ revenues. As mentioned above, our model predicts that firms that adjust their sourcing strategy to comply with RoO should experience a fall in global revenues. In line with these predictions, we show that constrained compliers — i.e., French firms that stop sourcing restricted inputs from non-TCA countries to retain duty-free access to the UK’s market — experienced a decrease in global revenues compared to unconstrained compliers — i.e., French firms that did not have to adjust their sourcing strategies to obtain origin on their final goods.

We also exploit the structure of our theoretical model to compute the expected cost of complying with RoO. Our estimates imply that compliance with TCA RoO increased the expected input sourcing costs of French firms by 9.7% on average.

Our paper is related to two main streams of literature. The first is the literature on the effects of regional trade agreements, which generally abstracts from the role of preferential RoO due to the difficulty in measuring these sourcing restrictions.⁶ This is for example the case of [Kehoe and Ruhl \(2013\)](#) and [Caliendo and Parro \(2015\)](#), which study the effects of NAFTA on trade and welfare.⁷ [Conconi et al. \(2018\)](#) show that accounting for RoO

⁶The literature on RoO is mostly theoretical in nature. Early work by [Grossman \(1981\)](#) investigates the effects of host government requirements that foreign firms use a certain proportion (measured by quantity or value) of host country inputs for their output to be sold in the host market. [Ju and Krishna \(2005\)](#) study firms’ incentives to comply with RoO in a setting in which firms are ex-ante identical, but may choose different equilibrium responses to the input requirements. [Head et al. \(2024\)](#) develop a model in which firms can source a continuum of inputs from both within and outside an FTA. They show that RoO can give rise to a Laffer curve, whereby stricter content requirements initially expand regional part sourcing, but contract it when set at levels above a threshold. [Ornelas and Turner \(2024\)](#) develop a property-rights model to study the effects of RoO on investments. They show that these rules can alleviate the hold-up problem when investments are non-contractible.

⁷[Kehoe and Ruhl \(2013\)](#) focus on changes in trade patterns driven by countries starting to export goods that they had not exported before. They find that the extensive margin is a crucial factor in explaining the increase in trade after trade liberalizations. On average, it accounts for 9.9 percent of the growth in trade for the NAFTA country pairs. [Caliendo and Parro \(2015\)](#) build on [Eaton and Kortum \(2002\)](#) to develop a

can reverse the conclusions about the effects of NAFTA: while [Caliendo and Parro \(2015\)](#) find little evidence of trade diversion and conclude that “the rest of the world was hardly affected by NAFTA” (p. 25), [Conconi et al. \(2018\)](#) show that NAFTA RoO on final goods reduced the growth rate of Mexican imports of intermediates from third countries relative to NAFTA partners. We provide the first firm-level evidence of the firm-level effects of RoO, by leveraging the introduction of these sourcing restrictions in post-Brexit EU-UK trade relations. A few recent studies examine the effects of the post-Brexit EU-UK FTA, abstracting from RoO. For example, [Freeman et al. \(2025\)](#) examine the impact of Brexit on UK’s trade with the EU relative to its trade with the rest of the world. Their results suggest that TCA firms did not switch away from non-TCA suppliers. In fact, “the new TCA trade relationship led to a sudden and persistent 25% fall in relative UK imports from the EU”. Our paper shows that TCA RoO distort French firms’ sourcing decisions, leading them to move away from non-TCA suppliers.

Another stream of literature emphasizes the productivity-enhancing effects of input tariff liberalization (e.g., [Amiti and Konings, 2007](#); [Goldberg et al., 2010](#); [Halpern et al., 2015](#); [Blaum et al., 2018](#)). Our paper shows that RoO act like input tariffs and have a “cascade effect”: the presence of these sourcing restrictions from RoO implies that preferential tariffs in FTAs on final goods are part of the implicit cost of importing intermediate goods from non-FTA countries. We show that, by distorting input sourcing decisions, these murky trade barriers increase firms’ costs, reducing their sales, employment, and productivity.

2 Brexit as a natural experiment

On 23 June 2016, the UK electorate was asked in a referendum whether the country should remain a member of the EU or withdraw from it — an option commonly termed as Brexit. The electorate voted 52% to 48% in favor of Brexit. The outcome was unexpected: in the twelve months before the referendum, the probability of a leave vote implied by prediction markets fluctuated between 25% and 35%, and never exceeded 45% ([Graziano et al., 2021](#)). The leave vote was a major political and economic shock, which led to a negative reaction of financial markets worldwide ([Breinlich et al., 2018](#)), and to resignation of UK prime minister David Cameron, who was succeeded by Theresa May three weeks after the vote.

In her Lancaster House Speech in January 2017, Theresa May formally outlined the Brexit goals of her government, explicitly ruling out membership in the EU’s Single Market and the Customs Union, to allow the UK to negotiate independent trade deals. In March 2017,

tractable model of tariff policy evaluation, which allows to decompose and quantify the differential role of intermediate goods and sectoral linkages.

the UK triggered Article 50, formally initiating the withdrawal process and establishing a two-year negotiation period. Still, the future UK-EU trade arrangement remained uncertain — with speculations ranging from some sort of Single Market and Customs Union membership, an FTA, and a “No-Deal” scenario. In November 2018, the UK and EU published a comprehensive draft Withdrawal Agreement, which outlined the basis for the new relationship. A deal was finally reached in December 2020, when the UK and the EU concluded the negotiations of an FTA, the TCA, which entered into force in January 2021.

During the transition period — after the Brexit referendum, but before the entry into force of the TCA — the UK was still part of the EU’s Customs Union and Single Market, so trade relations were unaffected. Agents expected trade barriers between the UK and the EU to increase, but there was great uncertainty over when any increase would occur and what form it would take (Bloom et al., 2018).⁸

In our paper, we exploit the entry into force of the TCA as a trade shock that led to the introduction of RoO in UK-EU trade relations: since January 2021, goods traded between the UK and the EU receive tariff-free treatment only if they comply with the RoO of the TCA. These rules could have been avoided and unconditional free trade could have been preserved had the UK decided to remain in the EU Customs Union, giving up the ability to negotiate independent trade deals.⁹ This is the first paper to code the RoO embedded in the text of the TCA and to theoretically and empirically study the firm-level effects of these sourcing restrictions.¹⁰

3 A Model of rules of origin

Rules of origin generate a trade-off for firms in an FTA like the TCA: firms that comply with RoO get preferential access to the market of the FTA partner(s), but must source certain within the FTA; on the other hand, firms that do not comply lose preferential access to the FTA partner(s), but are unrestricted in their sourcing strategies. To highlight this trade-off, we develop a model with heterogeneous firms that choose whether to comply with RoO sourcing restrictions on a composite intermediate input bundle to obtain preferential access for their final good. In line with our empirical analysis, we focus on changes in tariff classification requirements, which imply that specific inputs cannot be imported from third

⁸The existing literature on Brexit documents negative economic effects in the UK already during the transition period, e.g., increase in inflation and food prices (e.g., Breinlich et al., 2018; Bakker et al., 2022) and decrease in UK inward FDI (e.g., Serwicka and Tamberi, 2018; Breinlich et al., 2020).

⁹This is because RoO are meant to prevent trade deflection, which cannot arise in customs unions since members have the same external tariffs.

¹⁰Other papers study the effects of other frictions introduced by the TCA, including visa restrictions Brolli (2026) and border documentation Flach et al. (2026).

countries for an output to be considered as originating in the FTA.¹¹

3.1 Model setup

We consider a simple setup with a set $\mathcal{N} = \{H, F, R\}$ that includes three countries: home (H) and FTA partner (F) which form an FTA, and a rest of the world aggregate R . Before the FTA, flows between H and F were unconditionally duty free; under the FTA, duty-free treatment becomes conditional on satisfying restrictions on input sourcing. In our empirical setting, H maps to France and other EU-27 member states, F maps into the UK, and R represents all non-TCA countries.

The trade costs between countries H and F (i.e., τ_{HF} and τ_{FH}) capture preferential access between the two countries within the FTA area, which is assumed to be the same before and after the introduction of the FTA.

We assume that monopolistically competitive firms produce differentiated varieties of a final good i using labor and a composite intermediate input j . For simplicity, in what follows, we omit the subscripts i and j .

Firms located in a destination country $d \in \mathcal{N}$ have a common ex-ante distribution of the cost of sourcing the composite intermediate input from an origin country $o \in \mathcal{N}$. Each firm f draws a realization for its intermediate input cost. Thus, firms within country d are ex-ante identical, but ex-post heterogeneous in their intermediate input cost due to the realized draws from the distribution.

An implicit assumption is that firms cannot create production lines that are specific to the destination market of the final good: firm f located in d uses the same intermediate input j sourced in origin country o for the production of its final good i , regardless of the country to which that final good is exported. As shown in Section 3.8, this implies that RoO will distort firms' input sourcing decisions, with negative consequences on their sales in all destination markets.¹²

A firm f makes four decisions. First, it decides whether to set up production, paying a sunk entry cost. Second, conditional on entry, the firm draws the intermediate input cost. Third, depending on the observed draw from the cost of sourcing the intermediate input bundle, the firm chooses from which country to source the intermediate input. Finally, it chooses in which markets to sell its final good.

¹¹Changes in tariff classification rules are more restrictive and pervasive than value added rules, which require a minimum domestic cost share to obtain preferential access to the market of FTA partner(s). See [Head et al. \(2024\)](#) for a model with value-added sourcing restrictions.

¹²If we allowed for differentiated product lines by destination market, RoO would only affect the choice of inputs used to produce the final good that is exported to the FTA partner F , and only sales in this market would be negatively affected.

In the baseline version of the model, we consider single-output firms. The insights we derive in our baseline model about the decision of firms on whether to comply with sourcing restrictions carry over to a setting with multi-output firms under assumptions about the inputs used in production being output specific, and the firm making the decision to comply or not with sourcing restrictions being independent across outputs (i.e., a firm can decide to comply with restrictions on one output and to not comply on another output).

3.2 Input sourcing

Let $\tau_{od} \geq 1$ denote the full trade cost (comprehensive of both the iceberg trade cost) for delivering the intermediate input from origin country o to firms in destination country d .¹³

Each firm $f \in \mathcal{F}_d$ draws a realization for its intermediate input cost a_{od} from a Weibull distribution with shape parameter $\theta \geq 1$ that is common across countries, and country-specific scale $\mu_{od} > 0$. The shape parameter θ governs the dispersion of the sourcing cost draws, with larger values implying less heterogeneity across firms. The scale parameter μ_{od} affects the level of sourcing costs from o to d , with lower values shifting the distribution toward lower costs, leading to, on average, cheaper sourcing of inputs.

Then, for a firm f with a realized input cost a_{od} , the delivered input cost

$$z_{fod} = \tau_{od} a_{fod}.$$

Notice that z_{fod} is a random variable that follows a Weibull distribution with shape parameter θ and scale $\Lambda_{od} \equiv \tau_{od}\mu_{od}$.

The trade costs between countries H and F (i.e., τ_{HF} and τ_{FH}) capture preferential access between the two countries within the FTA area, which is assumed to be the same before and after the introduction of the FTA.¹⁴ This assumption is justified in our setting: before the entry into force of the TCA, the UK and the EU-27 were part of the EU Customs Union, in which qualifying for preferential tariff treatment was not conditional to compliance with sourcing restrictions.

Firms within the FTA decide whether to comply with RoO sourcing restrictions. We can thus distinguish two types of firms $g \in \{C, NC\}$, with C (NC) denoting a firm that

¹³We assume that $\tau_{dd} = 1$, i.e., domestic sourcing is costless. We then impose that $\tau_{HH} < \tau_{FH} < \tau_{RH}$ meaning that the bilateral trade cost of sourcing from the FTA partner is lower than from the rest of the world (capturing geographical proximity as well as the preferential import tariff). Country R is subject to the same trade cost, which includes MFN treatment, when delivering the input to H or F both before and after the introduction of the FTA between H and F .

¹⁴This assumption rules out cases in which a source country o is not the best supplier to a firm with a_{fod} before the FTA, but could become the best supplier after the FTA due to changes in trade costs (e.g., switching from MFN to preferential tariff rates).

complies (does not comply) with sourcing restrictions.

Firms that comply with sourcing restrictions choose to source the intermediate input from the lowest cost supplier country among the constrained set of suppliers (i.e., countries within the FTA area):

$$m_{fd}^C \equiv \min\{z_{fHd}, z_{fFd}\}, \quad d \in \{H, F\},$$

where m_{fd}^C is the (lowest) intermediate input cost for a complier firm f located in country d .

By contrast, a non-complier firm f considers country R outside of the FTA as possible supplier of the intermediate input, implying that the lowest intermediate input cost m_{fd}^{NC} is equal to

$$m_{fd}^{NC} \equiv \min\{z_{fHd}, z_{fFd}, z_{fRd}\}, \quad d \in \{H, F, R\}.$$

The lowest intermediate input cost m_{fd}^{NC} is also the solution to the sourcing problem for firms pre FTA since they faced no sourcing restrictions before the introduction of the FTA.

Notice that, if a country within the FTA area is the lowest cost supplier for a non-complier firm f in $o \in \{H, F\}$, then switching to the compliance regime is costless for that firm (i.e., $m_{fd}^C = m_{fd}^{NC}$).¹⁵ In other words, since the lowest cost supplier to the firm is within the FTA area the sourcing restrictions do not impose a change to a higher cost input supplier for the firm. On the other hand, firms for which country R is the lowest cost supplier face an increase to their intermediate input cost to comply with sourcing restrictions (i.e., $m_{fd}^C > m_{fd}^{NC}$).

For a complier firm, the probability to source the intermediate input from $o \in \{H, F\}$ is

$$\Pr(z_{fod} = m_{fd}^C) = \frac{(\tau_{od}\mu_{od})^{-\theta}}{\sum_{o \in \{H, F\}} (\tau_{od}\mu_{od})^{-\theta}}. \quad (1)$$

By contrast, for a non-complier firm, the probability to source the intermediate input from $o \in \{H, F, R\}$ is

$$\Pr(z_{fod} = m_{fd}^{NC}) = \frac{(\tau_{od}\mu_{od})^{-\theta}}{\sum_{o \in \{H, F, R\}} (\tau_{od}\mu_{od})^{-\theta}}. \quad (2)$$

Notice that the denominator in equation (2) is larger than in equation (1). Intuitively, having access to suppliers from a broader set of countries increases competition, leading to a lower probability that the input is sourced from a country $o \in \{H, F\}$.

¹⁵The set of supplier countries under the compliance regime is a subset of the set of countries in the non-compliance regime. Thus, if a country within the FTA area is a solution to the problem of the minimum delivered intermediate input cost in the broader set, then it is also the minimum in the restricted subset.

3.3 Production

Each firm $f \in \mathcal{F}_d$ located in country $d \in \{H, F, R\}$ produces a variety i of a final good using labor L and a composite intermediate input j according to a Cobb-Douglas technology

$$q_{fd} = L_{fd}^\beta j_{fd}^{1-\beta}, \quad 0 < \beta < 1. \quad (3)$$

We denote the wage with w_d . The marginal cost of the intermediate input varies by type of firm $g \in \{C, NC\}$

$$c_{fd}^g = b w_d^\beta (m_{fd}^g)^{1-\beta}, \quad b \equiv \beta^{-\beta} (1-\beta)^{-(1-\beta)}, \quad (4)$$

where $c_{fd}^C \geq c_{fd}^{NC}$ since $m_{fd}^C \geq m_{fd}^{NC}$. Notice that the compliance decision determines the firm-specific unit cost m_{fd}^g of the composite intermediate input, which in turn maps into the marginal cost c_{fd}^g through the Cobb-Douglas technology. The two minima, m_{fd}^C and m_{fd}^{NC} , differ whenever the compliance input cost is higher than the non-compliance one, i.e., when R is the lowest cost source of the unrestricted set of suppliers. The two minima are instead equal to each other when either H or F is the lowest cost source of the intermediate input for a firm f in $d \in \{H, F\}$. We define the compliance cost $\tilde{c}_{fd}$ as the ratio between the marginal cost of firm f if it complies with RoO sourcing restrictions and its marginal costs in case of non compliance:

$$\tilde{c}_{fd} \equiv \frac{c_{fd}^C}{c_{fd}^{NC}} = \left(\frac{m_{fd}^C}{m_{fd}^{NC}} \right)^{1-\beta} \geq 1. \quad (5)$$

3.4 Demand

Consider a representative consumer in country d with CES preferences over the set of differentiated final good varieties i

$$U_d = \left(\sum_{o \in \mathcal{N}} \sum_{g \in \mathcal{G}_o} \int_{f \in \mathcal{F}_o^g(d)} (q_{fod}^g)^{\frac{\sigma-1}{\sigma}} df \right)^{\frac{\sigma}{\sigma-1}}, \quad \sigma \equiv \frac{1}{1-\rho} > 1, \quad 0 < \rho < 1, \quad (6)$$

where σ denotes the elasticity of substitution across varieties. The set of available firm types is $\mathcal{G}_o = \{C, NC\}$ for $o \in \{H, F\}$ and $\mathcal{G}_o = \{NC\}$ for $o \in \{R\}$, since the rest of the world country does not face sourcing restrictions. We denote with $\mathcal{F}_o^g(d) \subseteq \mathcal{F}_o$ the set of firms of type g located in country o that sell their final good in destination d .¹⁶ Whether or not

¹⁶The set of available variates in market d is equal to $\Omega_d = \cup_{o \in \mathcal{N}} \cup_{g \in \mathcal{G}_o} \mathcal{F}_o^g(d)$.

firms in the FTA comply with RoO sourcing restrictions affects demand through the effect on the delivered price of the final good. The consumer in d maximizes consumption of a firm-specific variety subject to the budget constraint

$$\sum_{o \in \mathcal{N}} \sum_{g \in \mathcal{G}_o} \int_{f \in \mathcal{F}_o^g(d)} p_{fod}^g q_{fod}^g df = E_d, \quad (7)$$

where E_d denoted total expenditures in country d ; the delivered price in destination market d of the final good exported by firm f of type g located in country o is equal to p_{fod}^g . The CES price index in destination d is

$$P_d = \left(\sum_{o \in \mathcal{N}} \sum_{g \in \mathcal{G}_o} \int_{f \in \mathcal{F}_o^g(d)} (p_{fod}^g)^{1-\sigma} df \right)^{\frac{1}{1-\sigma}}. \quad (8)$$

Final good demand in country d for a firm f of type g located in origin o is

$$q_{fod}^g = E_d P_d^{\sigma-1} (p_{fod}^g)^{-\sigma}. \quad (9)$$

Revenues from market d for firm f of type g located in country o are equal to

$$r_{fod}^g = E_d P_d^{\sigma-1} (p_{fod}^g)^{1-\sigma}. \quad (10)$$

3.5 Final good pricing

Under monopolistic competition, firm f takes aggregate expenditures and the price-index as given when setting the price of its final good. The factory-gate price for a firm of type g is denoted with p_{fo}^g and is equal to a markup over the marginal cost:

$$p_{fo}^g = \frac{\sigma}{\sigma-1} c_{fo}^g = \frac{c_{fo}^g}{\rho},$$

where $\rho = \frac{\sigma-1}{\sigma}$. Then, the delivered price for the final good in market d is equal to

$$p_{fod}^g = T_{od}^g p_{fo}^g = \frac{T_{od}^g c_{fo}^g}{\rho}, \quad (11)$$

where T_{od}^g denotes the total trade cost to export final good produced by a firm of type g from origin o to destination d . Substituting the delivered price back in equations (9) and (10) shows that compliance decision affects f 's sales in a market through changes in marginal production cost and bilateral trade costs.

We assume that serving the domestic market is costless and that exports to R always receive MFN treatment.¹⁷ For firms located within the FTA, the benefit of complying with RoO sourcing restrictions is that they face lower tariffs when exporting their final good to the FTA partner's market (i.e., $T_{od}^{NC} \geq T_{od}^C > 1$ for $o \neq d$ and $o, d \in \{H, F\}$).

Crucially, equation (11) implies that a firm in H or F that adjusts its input sourcing to comply with the RoO restrictions of the FTA is subject to an increase in marginal cost (i.e., $\tilde{c}_{fo} > 1$) that increases delivered prices for its final good in *all markets* compared to pre-FTA prices.

3.6 Profits by market

All firms within origin o pay a fixed cost f_{od} for $o, d \in \mathcal{N}$ to serve destination d . Profits accrued to firm f of type g located in country o and exporting its final good to market d are equal to

$$\pi_{fod}^g = \frac{r_{fod}^g}{\sigma} - f_{od}. \quad (12)$$

Using equations (10) and (11), we derive the marginal cost cutoff for a firm in country o to sell its final good in market d (i.e., the cost cutoff for which $\pi_{fod}^g = 0$):

$$c_{od}^{g,*} = \frac{\rho P_d}{T_{od}^g} \left(\frac{E_d}{\sigma f_{od}} \right)^{\frac{1}{\sigma-1}}. \quad (13)$$

Notice that the cutoff is specific to a firm's type g , as it depends only on aggregate characteristics of the destination market and bilateral costs. The compliance type g affects the cutoff only through changes in bilateral trade costs T_{od}^g .

We define the set of profitable destination markets served by firm f of type g located in country o as

$$\mathcal{D}_{fo}^g \equiv \{d \in \mathcal{N} : c_{fo}^g \leq c_{od}^{g,*}\}.$$

Given our assumptions on bilateral trade costs for final goods across types of firms, we have that the marginal cost cutoff differs across types only when the destination market is an FTA partner. In other words, since $T_{oR}^C = T_{oR}^{NC}$ and serving the domestic market is costless, we have that $c_{oR}^{C,*} = c_{oR}^{NC,*} = c_{oR}^*$ and $c_{oo}^{C,*} = c_{oo}^{NC,*} = c_{oo}^*$, while $T_{od}^{NC} \geq T_{od}^C \implies c_{od}^{NC,*} \leq c_{od}^{C,*}$ for $o \neq d$ and $o, d \in \{H, F\}$. Intuitively, the marginal cost cutoff is lower when the firm is a non-complier and faces higher bilateral trade costs (i.e., MFN rather than preferential tariff)

¹⁷We have that $T_{oo}^C = T_{oo}^{NC} = 1$ for $o \in \{H, F\}$. For firms in $o \in \{H, F\}$ the bilateral trade cost to serve country R does not differ across firm's types C and NC , since the R always applies an MFN tariff to imports from H and F (i.e., $T_{oR}^C = T_{oR}^{NC} > 1$ for $o \in \{H, F\}$).

to export to the FTA partner.

3.7 Compliance decision

We can finally characterize the conditions under which a firm f located in country $o \in \{H, F\}$ chooses to comply with the RoO of the FTA.¹⁸ For ease of comparison, we focus here on firms that serve all destination markets under both compliance types (i.e., firms for which $\mathcal{D}_{fo}^C = \mathcal{D}_{fo}^{NC} = \mathcal{N}$).¹⁹

Before the FTA, a firm located in country $o \in \{H, F\}$ faces no sourcing restrictions, and its revenues in the FTA partner's market are given by

$$r_{fod} = E_d P_d^{\sigma-1} \left(\frac{T_{od}^C c_{fo}^{NC}}{\rho} \right)^{1-\sigma}, \quad o \neq d, \quad o, d \in \{H, F\}.$$

As previously mentioned, we assume the same preferential tariff T_{od}^C between H and F before and after the entry into force of the FTA (since members of the TCA were previously members of the EU Customs Union). The firm's total revenues pre FTA are equal to $r_{fo} = r_{fod} + r_{foo} + r_{foR}$. We define the share of country d in the firm's pre-FTA revenues as $\gamma_{fod} \equiv \frac{r_{fod}}{r_{fo}}$.

After the entry into force of the FTA, a firm located in H or F faces sourcing restrictions and chooses its compliance type g . Exports of a complier (non-complier) firm receive preferential (MFN) tariff treatment in the FTA partner's market. We define the preferential margin $\tilde{T}_{od}$ as the ratio between the MFN and preferential bilateral trade costs faced by a firm in the FTA partner's market:²⁰

$$\tilde{T}_{od} \equiv \frac{T_{od}^{NC}}{T_{od}^C} \geq 1. \quad (14)$$

Post FTA, total profits for a firm of type g are equal to

$$\Pi_{fo}^g = \sum_{d \in \mathcal{D}_{fo}^g} \pi_{fod}^g = \sum_{d \in \mathcal{D}_{fo}^g} \left[\frac{r_{fod}^g}{\sigma} - f_{od} \right]. \quad (15)$$

¹⁸Firms in country R are not in the FTA, so they do not choose between compliance or non-compliance.

¹⁹In general, RoO compliance can affect the set of markets served by a firm f located in country o ($\mathcal{D}_{fo}^g$): if the firm was optimally sourcing the intermediate input from R before the FTA, switching to a supplier from $o \in \{H, F\}$ to comply with the RoO of the FTA may lead the firm to stop serving some markets.

²⁰In this model we define the preference margin as the ratio of trade costs so its value is relative to the compliance trade costs at the denominator. Instead, the traditional preference margin often used in the empirical literature is the absolute difference between MFN and preferential tariff rates.

A firm will choose to comply if and only if

$$\Pi_{fo}^C \geq \Pi_{fo}^{NC}. \quad (16)$$

This inequality can be re-written as

$$\Pi_{fo}^C - \Pi_{fo}^{NC} = \frac{r_{fo}}{\sigma} \left[\tilde{c}_{fo}^{1-\sigma} - 1 + \gamma_{fod} \left(1 - \tilde{T}_{od}^{1-\sigma} \right) \right] \geq 0, \quad o \neq d, \quad o, d \in \{H, F\}. \quad (17)$$

Excluding the case in which the firm is indifferent between the two types (i.e., the inequality binds at zero), the condition for a firm to choose compliance over non-compliance can be written as

$$\gamma_{fod} \left(1 - \tilde{T}_{od}^{1-\sigma} \right) > 1 - \tilde{c}_{fo}^{1-\sigma}, \quad o \neq d, \quad o, d \in \{H, F\}. \quad (18)$$

The left-hand side of this condition captures the gains of complying with the RoO of the FTA: the firm's incentives to comply with the sourcing restrictions increase with the share of the FTA partner's market in f 's revenues before the agreement (γ_{fod}) and the preferential margin ($\tilde{T}_{od}$). The right-hand side of equation (18) captures instead the costs of complying with RoO, which increase with $\tilde{c}_{fo}$, the ratio between the marginal cost of firm f if it complies with RoO sourcing restrictions and its marginal costs in case of non compliance.

3.8 Testable implications

The model delivers clear predictions about the effects of RoO on firms' sourcing decisions. We can distinguish three categories of firms: constrained compliers, non-compliers, and unconstrained compliers.

Constrained compliers These are firms that source the intermediate input from R absent RoO restrictions, for which the gains of retaining duty-free access to the FTA partner market are large enough to justify switching to sourcing from H or F . In this case, $\tilde{c}_{fo} > 1$ and the firms satisfy the inequality condition of equation (18).

Non-compliers These are firms that source the intermediate input from R absent RoO restrictions, and continue to do so post FTA, since the compliance costs are larger than the benefits of retaining preferential access to the FTA partner's market. In this case, the firms do not satisfy the inequality from equation (17).

Unconstrained compliers These are firms that do not source the intermediate input from R pre FTA absent RoO restrictions, so they do not need to distort their input sourcing

to retain preferential access to the FTA partner’s market. For such firms, compliance is not costly: they can maintain preferential access to the FTA partner’s market without having to adjust their sourcing decisions. In this case, $\tilde{c}_{fo} = 1$ and the right-hand side of equation (18) is equal to zero.

When looking at the effects of the introduction of TCA RoO in 2021, our model delivers two sets of testable predictions for French firms. The first set concerns firms’ sourcing decisions. From equation (2) we know that, pre TCA, there is a positive probability of sourcing the intermediate input from the rest of the world. Post TCA, firms for which the inequality condition of equation (18) binds will stop sourcing the input from non-TCA countries to obtain origin on their final good. For constrained compliers, the probability to source from non-TCA countries will thus decrease from a positive value to zero, since RoO imply that R is no longer part of the set of available suppliers. Our model implies that TCA RoO should decrease the incentives of French firms to source inputs from outside the TCA, and this effect should increase with their compliance gains, captured by $\gamma_{fod} \left(1 - \tilde{T}_{od}^{1-\sigma}\right)$ in our model.

In line with these predictions, in Section 5 we show that the probability that French firms source intermediate inputs from non-TCA countries decreases significantly after the introduction of TCA RoO in 2021. This effect increases in the firms’ gains from complying with RoO. Placebo tests confirm that, when there are no compliance gains because the preference margin is zero (i.e., $\tilde{T}_{od} = 1$), TCA RoO have no impact on firms’ sourcing decisions.

The second set of predictions concerns firms’ revenues. Our model implies that, after the entry into force of the TCA, constrained compliers should experience a fall in global revenues equal to $r_{fo} (\tilde{c}_{fo}^{1-\sigma} - 1)$. The intuition is that a French firm that adjusts its input sourcing to retain preferential access to the UK market will see an increase in its marginal cost; in our model, this leads to higher delivered prices for the firm’s final good in *all markets* compared to pre-TCA prices.²¹ There should instead be no effect on global revenues of unconstrained compliers, which do not need to adjust their sourcing strategy to retain preferential treatment in the UK market. In the case of non-compliers, the negative effects on revenues should be restricted to the UK market, due to the increase from preferential to the MFN tariff; in other markets, there should be no effect on revenues, since production costs are not affected for these firms.

In line with these predictions, in Section 6 we show that constrained compliers — i.e., French firms that stop sourcing restricted inputs from non-TCA countries to comply with TCA RoO — experienced a decrease in global revenues compared to unconstrained compliers

²¹In our model setup, there is full pass-through of the increase in input cost to factory-gate prices.

— i.e., French firms that did not have to adjust their sourcing strategies to obtain origin on their final goods.²²

4 Data and variables

4.1 Rules of origin in the TCA

The RoO of the UK-EU TCA are contained in Annexes 2, 3, and 5 of the text of the agreement. We parse the text of the TCA describing the RoO applicable to each good.²³

Procedures to obtain TCA origin

Since January 2021, EU (UK) firms face MFN tariffs when exporting their products to the UK (EU), unless these comply with TCA RoO. Exporters claiming preferential tariff treatment self-declare the origin of their products by filling a Statement on Origin, based on information demonstrating compliance with TCA RoO. The exporter must hold evidence that the goods meet the relevant rules, including production records, bills of materials, cost calculations, evidence of manufacturing processes. Whether evidence is needed for all inputs depends on how the product qualifies for origin. If the product is “wholly obtained” (e.g., agricultural products grown entirely in the EU), extensive input tracing is usually unnecessary. If the product is manufactured using both originating and non-originating inputs, the exporter needs enough information to demonstrate compliance with the relevant product-specific rule.

Firms are responsible for the accuracy of their origin claim and must retain supporting records to demonstrate compliance with TCA RoO in case of an audit by UK or EU customs authorities. These authorities have the right to verify origin claims, requesting all the supporting documents from importers and exporters (e.g., supplier declarations, bills of materials, costing records, production records, and accounting data). They can deny preferential tariff treatment and recover duties if the origin claim cannot be substantiated. Firms can face financial consequences and penalties if they incorrectly declare that goods satisfy the TCA rules of origin.²⁴

²²We also compare constrained compliers with non-compliers. As predicted, RoO reduce total exports (excluding the UK) of firms that drop non-TCA suppliers to comply with TCA RoO compared to firms that do not adjust their sourcing strategies.

²³We use the PyMuPDF library in Python to parse the text of Annex 3 of the TCA.

²⁴Penalties are governed by the domestic customs laws of the UK or the relevant EU Member State. In the UK, HMRC can impose civil penalties for inaccurate customs declarations in cases involving carelessness or non-compliance. EU customs legislation allows authorities to impose administrative penalties under Member State law and pursue criminal penalties for intentional customs fraud.

Coding TCA RoO

We use the methodology of [Conconi et al. \(2018\)](#) to code CTC RoO. These rules define the origin of a good based on changes in the Harmonized System (HS) product classification codes (at the chapter, heading, or subheading level) with respect to its input. CTC rules imply that, for a final good to be eligible for preferential tariff treatment, the production or sourcing of some of its inputs must take place within the FTA.

We also code all the VA requirements, which limit how much foreign content (non-originating materials) can be used to qualify for preferential tariffs. VA requirements specify the maximum percentage of the value of inputs used in the production of a final good that can be sourced from outside the TCA. These rules provide producers with more flexibility on how to source inputs to comply with the rule.

In the TCA, CTC and VA requirements are often combined, out of 5,386 HS6 products, there are 5,074 final goods with CTC requirements, 2,780 of which have alternative VA requirements and 77 of which have additional VA requirements.²⁵ There are 115 final goods subject only to VA requirements. Table [A-1](#) shows that the use of alternative VA requirements is heterogeneous across industries, with certain industries almost completely subject to such requirements (e.g., Chemicals). Below we provide three examples to illustrate the RoO contained in the TCA.

Example 1: The rule applied to HS 64.06.20 (“Outer soles and heels, of rubber or plastics”) states “CTH”. This rule is a change to HS heading (under the acronym CTH), where a heading corresponds to a 4-digit code in the HS classification. In the case of HS 64.06.20, the relevant heading is HS 64.06 (“Parts of footwear (including uppers whether or not attached to soles other than outer soles); removable in-soles, heel cushions and similar articles; gaiters, leggings and similar articles, and parts thereof,”). The rule implies that all inputs within this heading cannot be sourced outside the TCA for good HS 64.06.20 to be considered as originating.

Example 2: The rule applied to all HS6 goods within HS 35.05 (“Dextrins and other modified starches”) states “CTH except from non-originating materials of heading 11.08”. As in the previous example, “CTH” refers to products within the same HS heading. Thus products within HS 35.05 cannot be produced with inputs within the same heading sourced outside the TCA. This rule includes an additional requirement “except from non-originating

²⁵For 197 products we find no CTC and/or VA requirements. As an example, consider HS heading 75.02 (“Unwrought nickel”) for which the RoO does not apply any restriction to inputs used: “Production from non-originating materials of any heading”.

materials of heading 11.08”, which restricts sourcing outside the TCA inputs within HS heading 11.08 (e.g., “starches; inulin”).

Example 3: The rule applied to all HS6 goods within HS 74.08 (“Copper wire”) states “CTH and MaxNOM 50% (EXW)”. As in the previous examples, “CTH” restricts the sourcing of inputs within the same HS heading as the final good to which the rule applies, i.e., HS heading 74.08. The second part of this rule (“and MaxNOM 50% (EXW)”) is an additional VA requirement. This means that producers also need the value of non-originating materials, which could be products outside HS 74.08, not to exceed 50% of the ex-works price of the final product.

We focus on CTC requirements. As discussed below, for each final good, this allows us to clearly trace the input-output linkages embedded in the rules: for every final good, we can trace the inputs subject to CTC requirements in the TCA; similarly, we can link every input to the final goods with RoO that impose restrictions on its sourcing. This is not possible for VA requirements, which do not specify the restricted inputs.

We construct the indicator variable RoO_{ij} , which is equal to 1 if there is a CTC requirement on final good i restricting the sourcing of input j . Going back to example 1 above, producers of “Outer soles and heels, of rubber or plastics” (HS 64.06.20) cannot source inputs within heading HS 64.06 — e.g., “Parts of footwear, uppers and parts thereof, other than stiffeners (HS 64.06.10) — from non-TCA countries if they want to obtain origin on their final good. In this example, the indicator variable RoO_{ij} is equal to 1 for the output-input pair HS 64.06.20 - HS 64.06.10.

Figures 1, 2 and 3 provide a graphical representation of the sourcing restrictions RoO_{ij} derived from the TCA. Outputs i are on the vertical axis, whereas inputs j are on the horizontal axis. Almost all intermediate goods have rules of origin associated to several outputs and most of them with outputs that fall into the same sector category.

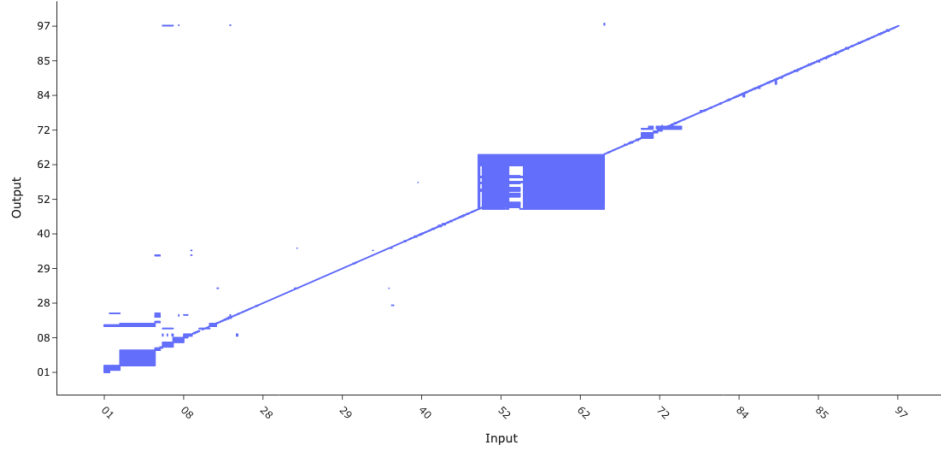
In robustness checks, we use data from input-output tables to restrict the attention to rules that apply to vertically related goods, i.e., the restricted good j is actually used as an input in the production of good i (see Section 5.3).

We can aggregate the RoO_{ij} sourcing restrictions at the input level to capture the number of final goods i that require input j :

$$RoO_j = \sum_i RoO_{ij}. \quad (19)$$

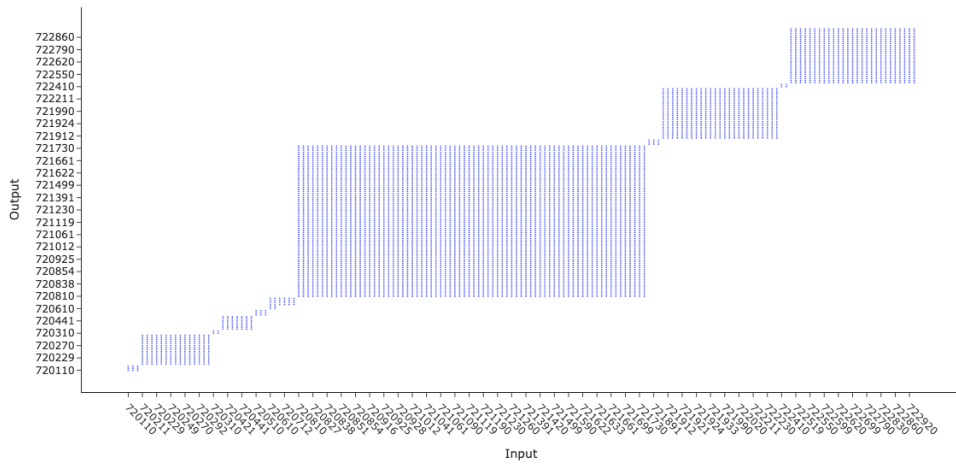
We can also aggregate at the output level to capture the number of restricted inputs j

Figure 1: Matrix of RoO_{ij} , all goods



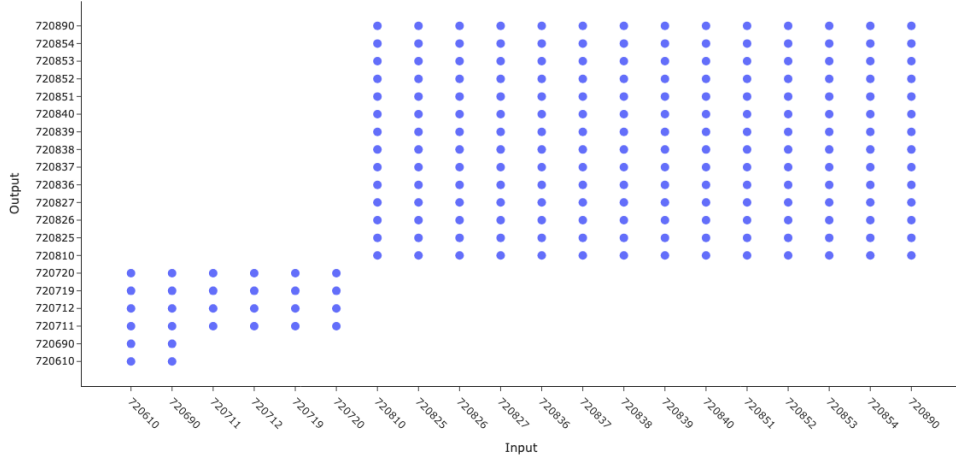
Notes: Blue dots represent output-input pairs with $RoO_{ij} = 1$, for all HS6 goods.

Figure 2: Matrix of RoO_{ij} , goods in chapter HS 72



Notes: Blue dots represent output-input pairs with $RoO_{ij} = 1$, for all goods within chapter HS 72.

Figure 3: Matrix of RoO_{ij} , goods in headings HS 72.06 to HS 72.08



Notes: Blue dots represent output-input pairs with $RoO_{ij} = 1$, for all goods within headings HS 72.06-72.08.

required by i 's RoO:

$$RoO_i = \sum_j RoO_{ij}. \quad (20)$$

Tables A-2 and A-3 show the heterogeneity of restrictions from RoO across HS Sections.

4.2 Tariffs

Tariffs data are from the MacMAP (CEPII) dataset, which provides information on MFN and preferential tariff rates at the country pair-product (HS6) level. The dataset covers 152 importing countries and 189 exporting countries in the period 2001-2022.²⁶

When looking at French firms' imports, we use data on EU *ad-valorem* tariffs to construct the variable $Applied\ Tariff_{jot}$, which measures the tariff faced by French producers importing intermediate input j from origin country o in year t . Table A-4 reports descriptive statistics of this variable by industry.

We also code the MFN tariff French firms face when exporting their final good i to the UK after 2021, if these goods do not comply with TCA RoO. The variable τ_i^{UK} is the MFN tariff rate applied by the UK after the entry into force of the TCA on imports of good i .²⁷ Table A-5 shows descriptive statistics of this variable by industry.

²⁶MacMAP provides tariff data on 3-year base for the years 2001, 2004, 2007, 2010, 2013, 2016, 2019, 2022. For the years in between (example 2021), we use values from the closest previous year to fill missing observations. MacMAP data are publicly available here: https://www.cepii.fr/cepii/en/bdd_modele/bdd_modele_item.asp?id=12.

²⁷We use data on UK MFN tariff rates in 2022 to construct this variable.

4.3 Administrative data for French firms

We use two main firm-level data sources to conduct our analysis: (i) customs records on the exports and imports of French firms; and (ii) annual account data for the universe of French firms (FICUS).

Firm-level data on the exports and imports of French firms are based on French Custom Records from the *Direction Générale des Douanes et des Droits Indirects* (DGDDI). We have information on import values (in euro) and quantities (volume in tons) of each French firm importing from a given origin, product (6-digit of the Harmonized System HS) in the period 2017-2024. Similarly, we have information on the exports of French firms to a given destination, product and year. This database is quasi-exhaustive of the universe of importing and exporting French firms.²⁸

Annual account data (FICUS) provide standard balance sheet information for the universe of French firms in the period 2017-2023. From this source we obtain information on firm characteristics (e.g., sales, employment sales per worker) used in the our empirical analysis.

4.4 Sample and firm-level variables

We include in our analysis all French firms that export to the UK at least once in the pre-TCA period (2017-2020). These are the firms that are potentially affected by the introduction of TCA RoO in 2021.²⁹

For each firm f , we use customs records to identify the relevant set of final goods I_f , i.e., all the HS6 products the firm exported to the UK at least once before the entry into force of the TCA. We also use customs records to identify the relevant set of intermediate inputs J_f , i.e., all the HS6 products imported by the firm at least once before the TCA. We denote with N_f^j the total number of inputs imported by the firm.³⁰

Combining our RoO_{ij} variable with information on the firm's final goods and inputs, we can create RoO firm-level measures. RoO_{fj} is the number of final goods produced by firm f with RoO restricting the sourcing of input j :

$$RoO_{fj} = \sum_{i \in I_f} RoO_{ij}. \quad (21)$$

²⁸Reporting is not mandatory for firms with trade values below 39,000 euros (within the EU destination) or 1,000 euros (extra-EU destinations). However, in practice many observations are below these thresholds.

²⁹The results hold if we exclude from the sample firms classified as wholesalers in the administrative data (those operating in NAF sector 46 "Commerce de gros").

³⁰We use the UN Broad Economic Categories (BEC) classification to identify HS6 codes associate with final goods and intermediate inputs. We restrict the analysis to input-origin combinations above the 1% of pre-TCA import threshold (i.e., origin country o accounted for at least 1% of French imports of input j).

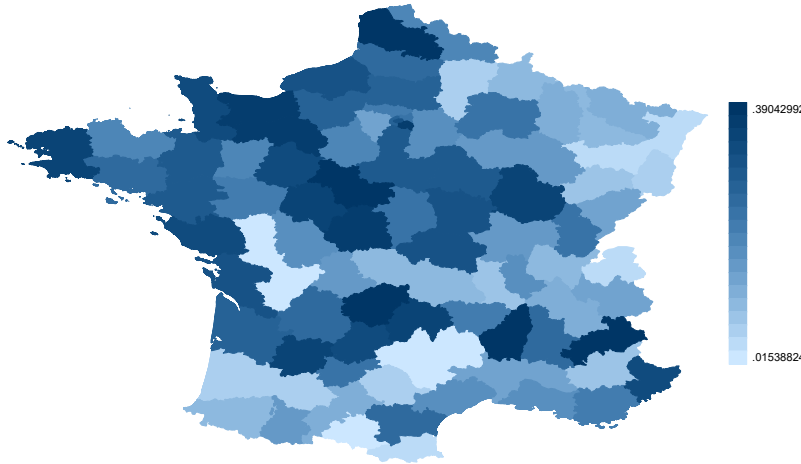
As RoO_{fj} increases, firm f produces more final goods that require input j originating within the TCA.

Combining data on firm-level imports with tariff data, we can construct a proxy for a firm's compliance gain. We first define the variable:

$$Share\ of\ Exports\ to\ UK_{fi} = \frac{\sum_{t < 2021} Exports_{fit}^{UK}}{\sum_{t < 2021} \sum_c Exports_{fit}^c}. \quad (22)$$

This is the pre-TCA share of firm f 's exports to the UK over total exports of product i (defined at the HS6 level). This variable is a proxy for γ_{fod} defined in Section 3.7, i.e., the share of the FTA partner's market in f 's revenues before the agreement.³¹ Figure 4 shows variation in the share of exports to UK across firms located in different regions of France.

Figure 4: Average of *Share of Exports to UK_{fi}* across firms in a region



The gains from complying with RoO sourcing restrictions are captured by the variable

$$Compliance\ Gains_{fi} = Share\ of\ Exports\ to\ UK_{fi} \times \tau_i^{UK}, \quad (23)$$

where τ_i^{UK} is the MFN tariff rate applied by UK after the entry into force of the TCA.

We also construct a firm-input specific measure of compliance gains. To this purpose, we

³¹In the model, γ_{fod} is defined using sales of a firm's final good in all markets. Administrative data for French firms (FICUS) do not provide product-level information on their sales. The annual production survey (EAP, Enquête Annuelle de Production) does contain information on domestic and export sales by product. However, the production survey covers only firms with at least 20 employees or 5 million euros in revenue. Moreover, the product classification used in the production survey does not match one-to-one with the HS6 classification used to define RoO in the TCA. We thus rely on product-level customs data to construct a proxy for γ_{fod} . As a result, *Share of Exports to UK_{fi}* does not include firm f 's sales in the domestic market.

define the subset I_{fj} of the firm's final goods that are subject to RoO that require input j (i.e., all final goods i in the set I_f for which $RoO_{ij} = 1$). The number of elements in the subset is defined as $N^{I_{fj}}$. Taking the average of *Compliance Gains* $_{fi}$ across all the outputs i in I_{fj} , we obtain

$$Compliance\ Gains_{fj} = \frac{\sum_{i \in I_{fj}} Compliance\ Gains_{fi}}{N^{I_{fj}}}. \quad (24)$$

5 Effects of RoO on firms' input sourcing

5.1 Empirical strategy

When examining the effects of RoO on firms' sourcing decisions, one may be concerned about the endogeneity of these rules. For example, during the negotiations of an FTA, some firms may lobby for stringent RoO, while others may push for more lenient rules. As a result, variation in the rules may be systematically related to firm characteristics (e.g., their industry, the extent to which they are declining).

We address this concern in several ways. First, notice that TCA RoO are mostly inherited from previous EU FTAs.³² Second, individual French firms are unlikely to shape the outcome of the TCA negotiations between the UK and the EU. To address any remaining concerns, we employ a triple-difference methodology, exploiting within-firm cross-product and cross-country variation in TCA RoO treatment over time. This allows us to include a set of fixed effects to address omitted variable concerns. Finally, we carry out placebo tests to verify that RoO have only effects on firms that have incentives to comply with the rules.

To examine whether TCA RoO distorted sourcing decisions of French firms, we estimate

$$\begin{aligned} Pr(Imports_{fjot}) = & \beta_0 + \beta_1 Non-TCA_{ot} \times Compliance\ Gains_{fj} + \\ & + \beta_2 Non-TCA_{ot} \times RoO_{fj} \times Compliance\ Gains_{fj} + \\ & \beta_3 Non-TCA_{ot} \times RoO_{fj} + \\ & \beta_4 Applied\ Tariff_{jot} + \\ & \delta_{fot} + \delta_{fjt} + \epsilon_{fjot}. \end{aligned} \quad (25)$$

The dependent variable is the probability that f imports intermediate input j from origin

³²For example, in the EU-Korea FTA, the RoO on HS 690911 ("Ceramic wares; for laboratory, chemical or other technical uses, of porcelain or china") is "Manufacture from materials of any heading, except that of the product"; this implies that, for the output to obtain origin, inputs within heading 6909 cannot be sourced outside the TCA. The corresponding rule in the TCA is "CTH", which restricts the sourcing of the same inputs (the EU simply switched from text to acronym).

country o in year t . RoO_{fj} is the number of firm f 's final goods imposing CTC RoO on input j . $Non-TCA_{ot}$ is an indicator identifying non-TCA countries in the post-TCA period (i.e., from 2021). $Compliance\ Gains_{fj}$ defined in equation (24) captures the firm's incentives to comply with TCA RoO on good j . $Applied\ Tariff_{jot}$ is the tariff applied by France on imports of good j from origin country o in year t . We include firm-origin-year (δ_{fot}) and firm-product-year (δ_{fjt}) fixed effects to address concerns about potential omitted variables.³³ We estimate (25) using a Poisson Pseudo-Maximum Likelihood (PPML).³⁴ We cluster standard errors at the firm level.

Recall from equation (24) that the variable $Compliance\ Gains_{fj}$ is defined only for the subset of firm's intermediate inputs that are subject to RoO. We thus restrict the analysis to inputs j for which RoO_{fj} is positive, and rescale this variable by subtracting the minimum so that $RoO_{fj} = 0$ when only one rule is imposed on the sourcing of the input. The β_1 coefficient thus captures the effects of TCA RoO when only one of the firm's final goods has a RoO restricting input j . The β_2 coefficient captures instead the effects of additional sourcing restrictions, i.e., the firm is producing multiple final goods that have RoO restricting the sourcing of input j ($RoO_{fj} > 0$).

Based on our theoretical model, we expect both β_1 and β_2 to be negative and significant, since the introduction of TCA RoO in 2021 reduced the incentives of French firms to source inputs from outside the TCA, in order to continue exporting their final goods duty free to the UK.

Our model also predicts that the β_3 coefficient should be insignificant, since it captures the effects of RoO for firms that have no incentives to comply with these rules (i.e., $Compliance\ Gains_{fj} = 0$). This is our placebo test. Finally, we expect $\beta_4 < 0$, reflecting the negative effects of input tariffs.

As explained in Section 4.1, there are 5,074 final goods with CTC requirements, 77 of which have additional VA requirements. To isolate the effects of CTC RoO, we can eliminate from the sample firms that are also constrained by additional VA requirements. To this purpose, we combine information on firm-level total production costs (from annual accounts) and import values (from customs records) with information on the specific threshold required by each VA requirement. We proceed in several steps:

1. For each firm f (producing final good i), we use pre-TCA data to construct the variables *Total Production Costs_f* (material inputs, intermediate inputs, labor costs) and

³³The qualitative results of our analysis are unaffected when we estimate specifications in which we also include product-origin-year fixed effects (δ_{jot}). In these specifications, the variable $Applied\ Tariff_{jot}$ is absorbed by the fixed effects, but we can still identify the effects of TCA RoO through the interaction variables $Non-TCA_{ot} \times Compliance\ Gains_{fj}$ and $Non-TCA_{ot} \times RoO_{fj} \times Compliance\ Gains_{fj}$.

³⁴We use the *ppmlhdf* command (Correia et al., 2020).

Imports from Non-TCA_f (value of imports from non-TCA countries). The variable *Non-TCA VA_f* = $\frac{\text{Imports from Non-TCA}_f}{\text{Total Production Costs}_f}$ captures the share of the firm’s VA originating outside the TCA.

2. For each final good i subject to VA RoO, we define the variable *Max VA_i*, which specifies the maximum percentage of value added accounted for by inputs from non-TCA countries for final good i to be originating.
3. We identify firms constrained by VA rules.³⁵ There are two cases:
 - *Non-TCA VA_f(i) ≤ Max VA_i ⇒ Constrained by VA_f = 0;*
 - *Non-TCA VA_f(i) > Max VA_i ⇒ Constrained by VA_f = 1.*

5.2 Results

Table 1 reports the results of estimating equation (25) using PPML. The estimates in column (1) are based on firms that face the CTC RoO contained in Annex 3 of the text of the TCA. As explained in Section 4.1, these rules impose sourcing restrictions on specific inputs and are sometimes combined with VA rules. To isolate the effects of CTC RoO, in column (2), we restrict the analysis to firms that are only constrained by CTC requirements, eliminating firms for which *Constrained by VA_f* = 1. In terms of sample, for each input j , we consider the same set of potential origin countries for all firms, i.e., all countries o that accounted for at least 1% of French imports of j before the TCA (see footnote 30).

In line with our model predictions, the coefficients of the interaction terms *Non-TCA_{ot} × Compliance Gains_{fj}* and *Non-TCA_{ot} × RoO_{fj} × Compliance Gains_{fj}* are always negative and significant, confirming that the introduction of TCA RoO reduced the probability that French firms source inputs from non-TCA countries. As shown in Table A-7, the results continue to hold even when adding product-country-year fixed effects. In this specification, the variable *Applied Tariff_{jot}* is absorbed by the fixed effects.

In terms of magnitude, the estimates of β_1 and β_2 in column (1) of Table 1 imply that, based on the average level of compliance gains, a one standard deviation increase in the number of RoO restricted inputs (i.e., *RoO_{fj}*) leads to a 3.2 percentage points decrease in the probability of importing from non-TCA, corresponding to a 29% reduction in the average probability of importing from non-TCA (9.6% in this sample). Based on the estimates

³⁵As an example, consider a firm f producing good i HS 63.07.20 (“Life jackets and life belts”). The RoO on this product (and all other products under heading HS 63.07) is “MaxNOM 40,” i.e., up to 40% of the value added can come from inputs from outside the TCA for the output to be considered as originating from the TCA and thus being eligible for preferential tariff treatment. Suppose that the *Non-TCA VA_f* is equal to 50%. In this case, the VA rule is binding, i.e., *Constrained by VA_f* = 1.

Table 1: Effects of TCA RoO on firms' imports of intermediate inputs
(all potential origin countries)

Dep var:	$Pr(Imports_{fjot})$	
	(1)	(2)
	All CTC rules	No VA rules
$Non-TCA_{ot} \times Compliance\ Gains_{fj}$	-3.034*** (1.067)	- 4.001*** (1.130)
$Non-TCA_{ot} \times RoO_{fj} \times Compliance\ Gains_{fj}$	-1.501* (0.785)	-1.823** (0.763)
$Non-TCA_{ot} \times RoO_{fj}$	0.012 (0.012)	0.014 (0.013)
$Applied\ Tariff_{jot}$	-0.839** (0.371)	-0.800* (0.409)
Estimator	PPML	PPML
Firm-Country-Year FE	Yes	Yes
Firm-Product-Year FE	Yes	Yes
Observations	4,983,209	4,431,340

Notes: The dependent variable is an indicator equal to 1 if firm f imports intermediate input j from origin country o in year t , and 0 otherwise. RoO_{fj} is the number of firm f 's final goods imposing CTC RoO on input j (in log). $Non-TCA_{ot}$ is an indicator identifying non-TCA countries in the post-TCA period (i.e., from 2021). $Compliance\ Gains_{fj}$, defined in equation (24), is included in logarithmic form as $\log(1 + Compliance\ Gains_{fj})$. $Applied\ Tariff_{jot}$ is the tariff applied by France on imports of good j from origin country o in year t and enters the specification as $\log(1 + Applied\ Tariff_{jot})$. The sample includes all origin countries o that accounted for at least 1% of pre-TCA imports of input j ; in column (1), it includes all firms that face the CTC RoO; in column (2), it excludes firms that are constrained by both CTC and VA sourcing requirements. Standard errors adjusted for clustering by firm in all specifications. ***, **, * significantly different from 0 at the 1%, 5%, and 10% levels respectively.

in column (2), a one standard deviation increase in the number of RoO restricted inputs leads to a 4.2 percentage points decrease in the probability of importing from non-TCA, corresponding to a 39% reduction in the average probability of importing from non-TCA.³⁶

As expected, the coefficient of $Non-TCA_{ot} \times RoO_{fj}$ is never significant, validating our placebo test: TCA RoO had no effect on input sourcing of French firms that had no incentives to comply with these rules (i.e., $Compliance\ Gains_{fj} = 0$).

Finally, the coefficient of $Applied\ Tariff_{jot}$ is always negative and significant, indicating that tariffs deter firms from sourcing inputs from foreign suppliers. In terms of magnitude, a one percent increase in import tariffs reduces the probability of importing an input from a given origin by 8 percentage points.

As mentioned before, the results of Table 1 include the same set of *potential* origin countries for all French firms. We next restrict the sample to the set of *actual* suppliers of each firm, i.e., those origin countries o from which a firm f was importing input j before

³⁶The magnitudes of the effects are based on in-sample descriptive statistics reported in Table A-6.

the entry into force of the TCA. The results are reported in Table 2. We follow the same structure as in Table 1: in column (1), we include all firms subject to CTC RoO; in column (2), we eliminate firms that are also restricted by VA requirements.

Table 2: Effects of TCA RoO on firms' imports of intermediate inputs
(only actual pre-TCA origin countries)

Dep var:	$Pr(Imports_{fjot})$	
	(1)	(2)
	All CTC rules	No VA rules
$Non-TCA_{ot} \times Compliance\ Gains_{fj}$	-1.155 (1.143)	- 1.153 (1.197)
$Non-TCA_{ot} \times RoO_{fj} \times Compliance\ Gains_{fj}$	-2.290*** (0.830)	-2.929*** (0.937)
$Non-TCA_{ot} \times RoO_{fj}$	0.014 (0.009)	0.014 (0.009)
$Applied\ Tariff_{jot}$	-0.047 (0.240)	-0.121 (0.263)
Estimator	PPML	PPML
Firm-Country-Year FE	Yes	Yes
Firm-Product-Year FE	Yes	Yes
Observations	1,501,847	1,322,951

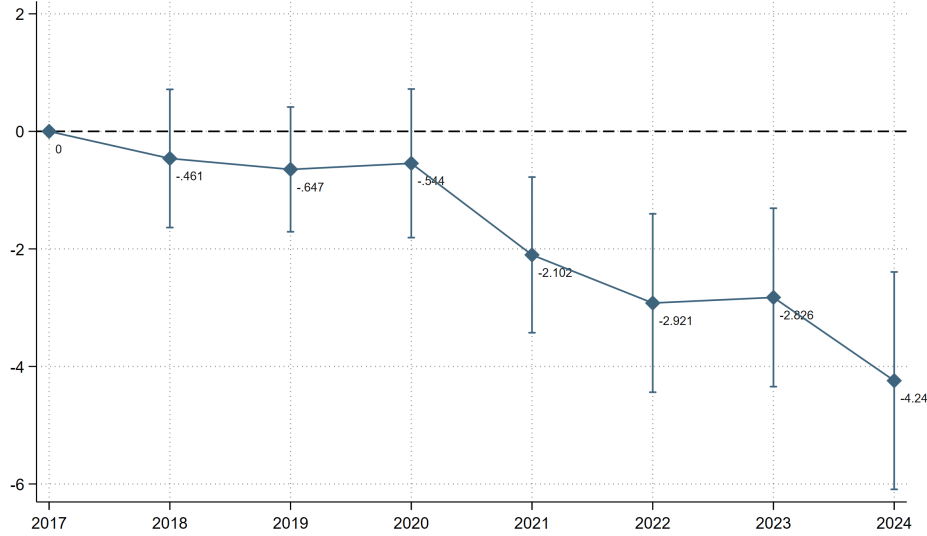
Notes: The dependent variable is an indicator equal to 1 if firm f imports intermediate input j from origin country o in year t , and 0 otherwise. RoO_{fj} is the number of firm f 's final goods imposing CTC RoO on input j . $Non-TCA_{ot}$ is an indicator identifying non-TCA countries in the post-TCA period (i.e., from 2021). $Compliance\ Gains_{fj}$, defined in equation (24), is included in logarithmic form as $\log(1 + Compliance\ Gains_{fj})$. $Applied\ Tariff_{jot}$ is the tariff applied by France on imports of good j from origin country o in year t and enters the specification as $\log(1 + Applied\ Tariff_{jot})$. The sample is restricted to origin countries o from which firm f was sourcing input j pre-TCA; in column (1), it includes all firms that face the CTC RoO; in column (2), it excludes firms that are constrained by both CTC and VA sourcing requirements. Standard errors adjusted for clustering by firm in all specifications. ***, **, * significantly different from 0 at the 1%, 5%, and 10% levels respectively.

The results confirm that French firms are more likely to stop sourcing inputs from their non-TCA suppliers after the introduction of TCA RoO in 2021. Notice that only the coefficient of the triple interaction $Non-TCA_{ot} \times RoO_{fj} \times Compliance\ Gains_{fj}$ is negative and significant, indicating that the trade diverting effects of RoO are driven by sourcing restrictions that apply to multiple final goods of the firm. In terms of magnitude, the coefficient of this variable in column (1) (column (2)) implies that one standard deviation increase in RoO leads to a 1.4 (1.8) percentage points decrease in the probability of importing from non-TCA, corresponding to a 6% (8%) reduction in the average probability of importing from non-TCA (25% in this sample). Notice that the coefficient of $Applied\ Tariff_{jot}$ loses significance when restricting the analysis to the reduced sample of origin countries.

We have also estimated an event-study specification of equation (25). Figure 5 plots the coefficients of $Non-TCA_{ot} \times RoO_{fj} \times Compliance\ Gains_{fj}$ when estimating the event study

specification of column (2) of Table 2. The estimates are negative and significant after 2021, confirming that the introduction of TCA RoO led French firms to stop sourcing inputs from non-TCA countries. The coefficients become increasingly large, suggesting that it takes time for firms to switch input suppliers. There are no significant pre-trends, indicating that firms did not adjust their sourcing decisions before 2021 in the anticipation of TCA RoO.

Figure 5: Event study estimates



Notes: the figure reports the coefficients of $Non-TCA_{ot} \times RoO_{fj} \times Compliance\ Gains_{fj}$ when estimating an event study specification of equation (25) based on the sample in column (2) of Table 2. Standard errors adjusted for clustering by firm.

In terms of magnitude, the coefficient for 2024 implies that, for a firm with average compliance gains, a one standard deviation increase in the number of RoO restricted inputs leads to a 2.6 percentage point reduction in the probability of sourcing from non-TCA countries, which corresponds to a 10% decrease in the average probability of importing from non-TCA (25% in this sample).

5.3 Additional robustness checks

The results of Table 1 show that the introduction of TCA RoO distorted French firms' sourcing decisions, decreasing the probability that they import restricted inputs from non-TCA countries. The results are robust to constructing the treatment variable RoO_{fj} using all CTC rules or only those that are not combined with VA rules (see columns 1 and 2), using different samples of origin countries (see Table 2), and further including product-country-year fixed effects (see Table A-7). In what follows, we report the results of a series of additional

robustness checks.

Excluding rules that do not apply to vertically related goods

Rules of origin should only affect firms' sourcing decisions if they apply to vertically-related goods. We verify that our results are robust to eliminating requirements RoO_{ij} in which good j is not actually used as an input in the production of good i .

To identify vertical linkages between products, we use detailed Input-Output (IO) tables from the US Bureau of Economic Analysis (BEA).³⁷ These include observations at the industry level based on the classification published by the BEA. We follow several steps to match these data to TCA RoO, which are defined at the HS6 level. First, we harmonize the BEA classification with the North American Industry Classification System (NAICS) 2017 using the concordance table provided by the BEA. Second, we use the crosswalk from [Pierce and Schott \(2012\)](#) between the NAICS 2017 classification and the HS 2017 classification.³⁸

Table A-8 shows that the results of Table 1 are robust to constructing the treatment variable RoO_{fj} using only rules for which $dr_{ij} > 0$.³⁹

Excluding the UK from set of origin countries

Before January 2021, when the UK was in the EU Customs Union, French imports from the UK were recorded via the Declaration of Trade in Goods (DEB). Businesses only had to file the detailed DEB statistical survey if their total French imports from other EU Member States exceeded 460,000 euros in a calendar year.⁴⁰ After the entry into force of the TCA in January 2021, French imports from the UK are no longer treated as intra-EU flows. Instead, they are recorded as standard third-country imports, meaning commercial goods require formal customs declarations regardless of value. There is no minimum threshold for filing a customs declaration, so all cross-border movements between the EU and the UK involve full customs declarations.

One might be concerned that changes in the data recording process may lead to artificial switches in the dependent variable in equation (25). In particular, a firm f may have already been importing input j from the UK, but pre TCA the variable $Pr(Imports_{fjot})$ could be coded as 0 for the UK as origin country if firm f was below the DEB reporting threshold.

³⁷We use the direct requirement matrix provided in the 2017 version of the IO tables.

³⁸The mapping from 4-digit industry codes to HS6 products suffers from measurement error: each industry code is associated on average to multiple HS 6-digit goods; there are also a few instances in which a HS-6 good is associated to more than one industry; moreover, not all HS6 products appear in the crosswalks between industry and product codes, which leads to fewer HS6 products being included in our analysis.

³⁹Direct requirement coefficients measure the amount of an input purchased to produce a dollar of industry output.

⁴⁰See <https://www.eurotax.fr/en/intrastat-emebi-ex-deb-in-france-2022-thresholds-in-europe/>.

Table A-9 show that the results are unaffected if we exclude the UK from the sample of origin countries.

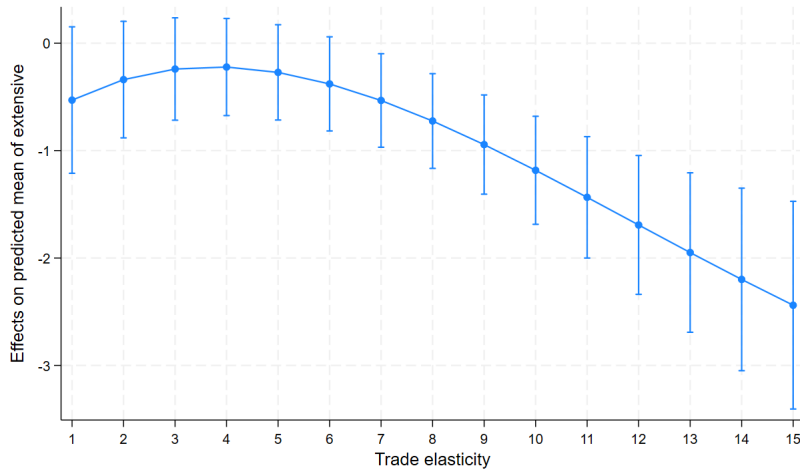
5.4 Heterogeneous effects

We next examine whether the effects of TCA RoO on firms sourcing restrictions depend on the substitutability of inputs and the number of inputs restricted by each rule.

Input substitutability

In our model, firms produce a variety of a final good using only labor and an intermediate input j restricted by RoO. In our data, firms may face RoO sourcing restrictions on several inputs, which may differ in their substitutability across countries. We would expect larger negative effects of RoO on non-TCA imports for inputs that are more substitutable. To verify this, we use data on the elasticity of substitution of HS6 products across countries from Fontagné et al. (2022). Figure 6 reports the coefficients of $Non-TCA_{ot} \times RoO_{fj} \times Compliance\ Gains_{fj}$ by values of elasticity of substitution of j , when estimating equation (25) using PPML. The figure confirms that, when faced by TCA RoO, firms are more likely to stop sourcing inputs from non-TCA countries when the inputs have a higher elasticity of substitution.

Figure 6: Heterogeneous effects by elasticity of substitution of inputs

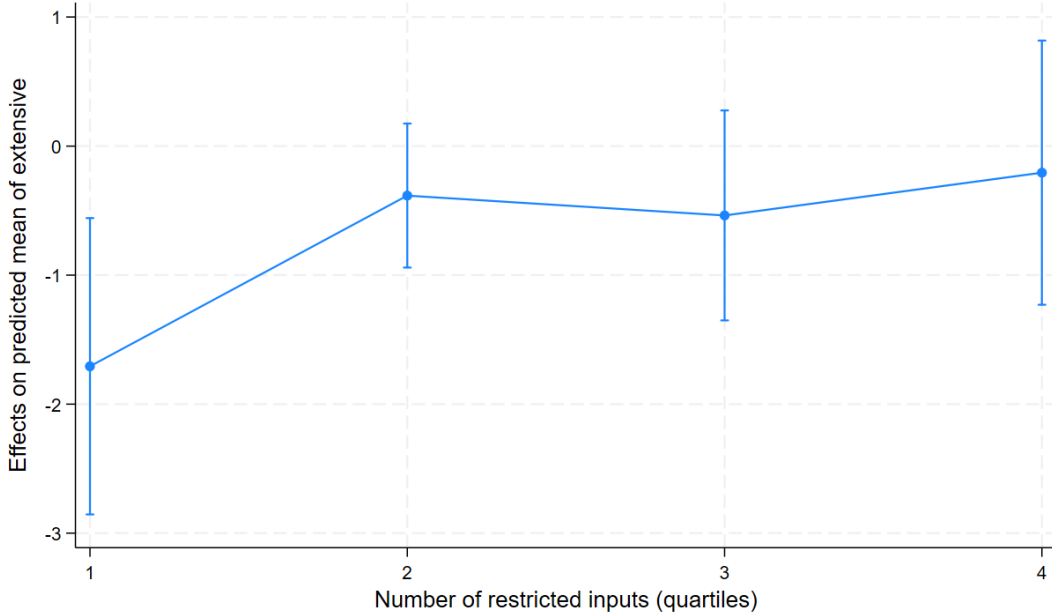


Notes: The figure reports the coefficients of $Non-TCA_{ot} \times RoO_{fj} \times Compliance\ Gains_{fj}$ by values of elasticity of substitution of j , when estimating equation (25) using PPML. Data on elasticity of substitution (in absolute value) from Fontagné et al. (2022). Standard errors adjusted for clustering by firm.

Number of inputs restricted by a RoO

In our model, RoO on final good i restrict only the sourcing of intermediate input j . In our data, the same rule may impose restrictions on several inputs (see Figure 3). Consider an hypothetical example of a RoO on a final good (female dresses) that imposes restrictions on the sourcing of two inputs (cotton and silk). A French producer that was importing these inputs from non-TCA countries before 2021 will have to switch to TCA suppliers for both inputs to retain duty free access to the UK for its final good after 2021. Notice that the compliance gains are constant while the compliance costs increase in the number of restricted inputs. We would expect no compliance — and thus no negative effects on imports from non-TCA countries — when RoO restrict too many inputs.

Figure 7: Heterogeneous effects by number of RoO restricted inputs



Notes: The figure reports the coefficients of $Non-TCA_{ot} \times RoO_{fj} \times Compliance\ Gains_{fj}$ by quartile in the number of RoO restricted inputs for firm f , when estimating equation (25) using PPML. Standard errors adjusted for clustering by firm.

Figure 7 reports the coefficients of $Non-TCA_{ot} \times RoO_{fj} \times Compliance\ Gains_{fj}$, for different quartiles of the number of inputs restricted by the same rule. The figure confirms that TCA RoO reduce the probability of sourcing from non-TCA countries, but only when the rules restrict a limited number of inputs: only the first quartile (which captures the effect of RoO restricting less than 4.25 inputs) is negative and significant.

6 Effects of RoO on firm-level outcomes

We next examine the effects of RoO on different firm-level outcomes. As discussed in Section 3.8, our model predicts that firms that adjust their sourcing strategy to comply with RoO should experience a fall in global revenues. There should instead be no effect on revenues of firms that do not need to adjust their input sourcing to retain preferential treatment in the UK market.

To assess the validity of this predictions, we first identify three sets of firms in our data:

- *Constrained Complier_f*: indicator variable equal to 1 if firm f imports at least one of the inputs restricted by TCA RoO from non-TCA countries before 2021, and does not import any of the restricted inputs from non-TCA countries after 2021;
- *Unconstrained Complier_f*: indicator variable equal to 1 if firm f does not import from non-TCA countries any input restricted by TCA RoO before and after 2021;
- *Not Complier_f*: indicator variable equal to 1 if firm f imports from non-TCA countries at least one input restricted by TCA RoO before and after 2021.

We then estimate the following model by Ordinary Least Squares (OLS):

$$Sales_{ft} = \beta_0 + \beta_1 Post-TCA_t \times Constrained\ Complier_f + \delta_f + \delta_t + \epsilon_{ft}, \quad (26)$$

where $Sales_{ft}$ are total sales, including to the domestic market, of firm f in year t . We include firm (δ_f) and year (δ_t) fixed effects to control for unobserved firms characteristics and time varying effects common across firms. We cluster standard errors at the firm level. The sample includes only constrained and unconstrained compliers. This restriction allows us to identify the effects of TCA RoO by comparing firms that had to adjust their sourcing decisions to comply with TCA RoO with firms that could obtain origin without adjusting their input sourcing.

Column 1 of Table 3 reports the results of estimating equation (26). In line with our model's predictions, the β_1 coefficient is negative and significant. This result confirms that, by distorting input sourcing decisions, RoO lead to a reduction in firms' global sales.

In the remaining columns of Table 3 we modify the dependent variable in equation (26) to examine the effects of RoO on other firm-level outcomes: size (captured by employment) and productivity (captured by sales per worker). All the coefficients are negative and significant, indicating that firms that had to adjust their sourcing decisions to comply with TCA RoO suffered a reduction in their employment and productivity.

Table 3: Effects of TCA RoO on firms' sales, employment, and productivity

Dep var:	Sales	Employment	Sales p/w
	(1)	(2)	(3)
$Post-TCA_t \times Constrained\ Complier_f$	-0.131*** (0.020)	-0.084*** (0.015)	-0.047*** (0.017)
Estimator	OLS	OLS	OLS
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	22,164	22,164	22,164

Notes: This table reports the results of estimating equation (26) on the sample of constrained and unconstrained compliers. In column 1 the dependent variable is a firm's total sales, $Sales_{ft}$. In the remaining columns we replace this variable with $Employment_{ft}$ and $Sales_{ft}/Employment_{ft}$. Standard errors in parentheses clustered at the firm level. ***, **, * significantly different from 0 at the 1%, 5%, and 10% levels respectively.

We can also compare constrained compliers with non-compliers. Our model predicts that constrained compliers should experience a fall in revenues in all markets, while non-compliers should only experience a decline in sales in the UK market, due to the loss in preferential tariff treatment. In line with this prediction, we find that total exports (excluding exports to the UK) of constrained compliers declined post TCA compared to non-compliers.⁴¹

7 Estimating of the expected costs of RoO compliance

In this section, we benefit from the structure of our theoretical model to infer the expected cost of complying with TCA RoO. Namely, in our model, firms make two choices. First, they decide whether to comply or not with RoO (C or NC) based on the share of UK market in firm's revenues (γ_{fod}), the preferential margin ($\tilde{T}_{od}$) and the ratio between the marginal cost of complying *vs* not-complying with RoO sourcing restrictions – see eq. (17). Second, firms decide from which country to source the intermediate input – see eq. (1)-(2). Hence, for a firm f , the probability of choosing country o to source the intermediate input can be decomposed as the product of the probability of complying *vs* non-complying with RoO ($\Pr(g)_f$) and the probability of sourcing from o conditional on the compliance decision ($\Pr(o|g)_f$):

$$\Pr(o)_f = \Pr(g)_f \times \Pr(o|g)_f, \quad \text{with } g \in \{C, NC\}. \quad (27)$$

Recall from equation (2) that under non-compliance the probability to source $o \in \{H, F, R\}$

⁴¹The coefficient of $Post-TCA_t \times Constrained\ Complier_f$ is negative and significant, i.e., -0.120 (0.056). Results are available upon request.

is the cheapest supplier to the firm in location d is

$$\Pr(o|NC)_f = \Pr(z_{fod} = m_{fd}^{NC}) = \frac{(\tau_{jod}\mu_{od})^{-\theta}}{\Phi_d^{NC}}, \quad \text{with} \quad \Phi_d^{NC} \equiv \sum_{c \in \{H, F, R\}} (\tau_{jcd}\mu_{cd})^{-\theta}. \quad (28)$$

Similarly, from equation (1) we know that, conditional on complying, the probability to source from $o \in \{H, F\}$ is

$$\Pr(o|C)_f = \Pr(z_{fod} = m_{fd}^C) = \frac{(\tau_{jod}\mu_{od})^{-\theta}}{\Phi_d^C}, \quad \text{with} \quad \Phi_d^C \equiv \sum_{c \in \{H, F\}} (\tau_{jcd}\mu_{cd})^{-\theta}. \quad (29)$$

Those conditional probabilities can be estimated from the following sourcing regression:

$$\Pr(z_{fod} = m_{fd}^g) = \exp(-\theta \log(\tau_{od}\mu_{od}) + \text{FE}_d^g), \quad \text{with} \quad \text{FE}_d^g = -\log \Phi_d^g. \quad (30)$$

Because we observe tariffs, we can estimate θ . The upper level choice is driven by the respective probabilities of choosing to comply or not. Using equation (18), we can write

$$\Pr(C)_f = \Pr[\gamma_{fod}(1 - \tilde{\tau}_{iod}^{1-\sigma}) > 1 - \tilde{c}_{fo}^{1-\sigma}]. \quad (31)$$

The term $\tilde{c}_{fo}$ is a ratio between the cost of complying vs non-complying for firm f . At the first stage, the firm does not know which precise country will be chosen, hence it must form expectation on the marginal cost of production in each case. On top of that, each of those costs has a random term. Given the appropriate (GEV) distribution on this random term, the expected cost is given by the Inclusive Value of choosing D : $-\frac{1}{\theta} \log \Phi_d^g$. This can be recovered from the lower stage of the estimation.

7.1 Empirical implementation

In our empirical analysis, we drop the subscript d (all firms in our sample are in France) and include j and t subscripts (to account for the fact that firms import multiple inputs j over several years t). Focusing on the pre-TCA period, when firms were not constrained by TCA RoO, we estimate the probability of a firm f importing input j from origin o in year t :

$$\Pr(\text{Imports}_{fjot} > 0) = \beta_1 \log(1 + \text{tar}_{jot}) + \delta_{ot} + \delta_{fjt} + \epsilon_{fjot}. \quad (32)$$

The variable of interest is $\log(1 + \text{tar}_{jot})$, the tariff paid by firm f when importing j from o in a given year. Note that since all importers are based in France, the δ_{ot} fixed effect has interpretation $-\theta \log(\mu_{ot})$, and μ_{ot} accounts for both production costs at origin and

other sources of trade costs (e.g., distance, language). The structural interpretation of our estimated β_1 is $-\theta$. By the logic of the conditional logit formulation of the choice (maintained under the PPML estimation), the chooser fixed effects correspond to:

$$\delta_{fjt} = -\log \sum_{o(j) \in \{H, F, R\}} \exp [\beta_1 \log(1 + \text{tar}_{jot}) + \delta_{ot}].$$

Note that we use the notation $o(j)$ to denote the set of origin countries available to *all* firms when importing j . This set is constructed such that a country of origin represents at least 1% of French imports of that input in the pre-TCA sample (as in Table 1). Therefore all f importing j have the same number of potential origins and therefore the same estimated fixed effect δ_{fjt} . Finally, since the regression is run pre-TCA, the set is unconstrained, that is includes potentially all countries in the world ($o(j) \in \{H, F, R\}$ in the model).

We estimate (32) with PPML on the pre-TCA period (2017-2020) for all French importers. The resulting estimate is $\hat{\beta}_1 = -3.582$, with standard error of 0.275, as shown in Table 4.⁴²

Table 4: The elasticity of import probability to tariffs

Dep var:	$Pr(Imports_{fjot})$
<i>Applied Tariff</i> _{jot}	-3.582*** (0.275)
Estimator	PPML
Firm-Product-Year FE	Yes
Origin-Year FE	Yes
Observations	4,785,688

Notes: The dependent variable is an indicator equal to 1 if firm f imports intermediate input j from origin country o in year t , and 0 otherwise. *Applied Tariff*_{jot} is the tariff applied by France on imports of good j from origin country o in year t and enters the specification as $\log(1 + \text{Applied Tariff}_{jot})$. The sample includes all origin countries o that accounted for at least 1% of pre-TCA imports of input j and covers the pre-TCA period 2017-2020. Standard errors adjusted for clustering by origin-product-year. ***, **, * significantly different from 0 at the 1%, 5%, and 10% levels respectively.

Using the estimates of regression (32), we compute the expected costs of RoO for constrained compliers. Intuitively, the exercise we carry out is the following: take *future* constrained compliers before the entry into force of the TCA; imagine that they would have been constrained at the time, what is their new expected sourcing costs? To address this question, we proceed in four steps:

1. We obtain $[(1 + \text{tar}_{jot})\mu_{ot}]^{-\hat{\theta}}$ using $\exp[-\theta(1 + \text{tar}_{jot}) + \delta_{ot}]$.

⁴²This tariff elasticity of sourcing decision is lower than the ones estimated for battery sourcing in the EV industry (around -8) by Head et al. (2026), but larger than the preferred sourcing elasticity of $\hat{\theta} = 1.8$ by Antràs et al. (2017).

2. For each firm-input-year, we sum twice across the relevant set of origins, using first the unconstrained and then constrained sets of origins. This gives us

$$\mathbb{I}_{jt}^{\text{NC}} = \log \sum_{o(j) \in \{H, F, R\}} [(1 + \text{tar}_{jot}) \mu_{ot}]^{-\hat{\theta}},$$

$$\mathbb{I}_{jt}^{\text{C}} = \log \sum_{o(j) \in \{H, F\}} [(1 + \text{tar}_{jot}) \mu_{ot}]^{-\hat{\theta}}.$$

3. For a given firm-section combination f (a section being the most aggregate level of the HS classification), we compute a weighted average across inputs, using the inputs' share ω_{jt} in the firm's total cost as weights.

$$\mathbb{I}_{ft}^{\text{NC}} = -\frac{1}{\hat{\theta}} \omega_{jt} \sum_{j(f)} \mathbb{I}_{jt}^{\text{NC}}, \quad \text{and} \quad \mathbb{I}_{ft}^{\text{C}} = -\frac{1}{\hat{\theta}} \omega_{jt} \sum_{j(f)} \mathbb{I}_{jt}^{\text{C}}$$

4. Finally, we average firms-sections combinations to obtain average expected sourcing cost under compliance and non-compliance.

Column (1) of the Table 5 reports the expected increase in input sourcing costs for firms in each HS section.⁴³ On average, expected input sourcing costs increased by 9.7% for firms complying with TCA RoO. Column (2) of the same Table reports the share of inputs subject to TCA RoO that were imported from non-TCA countries before 2021. On average, 18.8% of imported inputs were sourced from third countries in the pre-TCA period. Interestingly, some of the sectors with the highest increase in expected input sourcing costs (e.g., minerals, footwear) are also those that feature higher-than-average import shares from non-TCA countries. Figure 8 presents a binned scatter plot showing the correlation between the firm-level expected increase in input sourcing costs and the firm-level share of inputs imported from non-TCA countries.

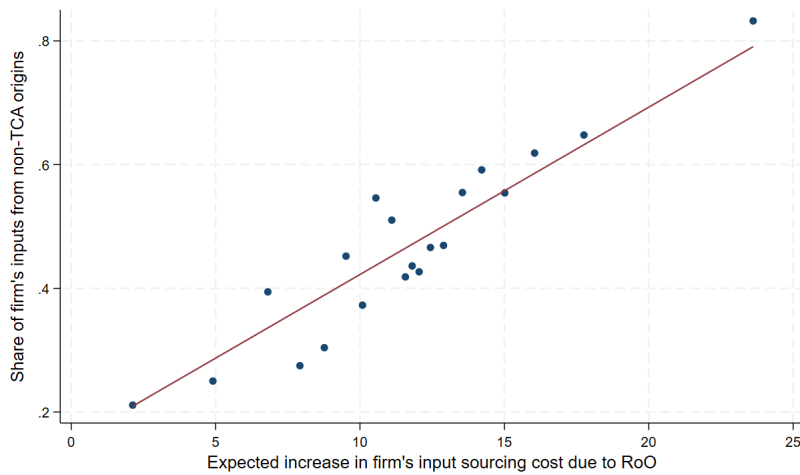
⁴³There are no statistics for two sectors: Chapter 93 ("Arms and Ammunition") because inputs in this chapter are not subject to CTC requirements; and Chapter 97 ("Works of Art"), because it does not include intermediate inputs based on the BEC classification.

Table 5: Expected increase in input sourcing costs due to RoO compliance

HS Chapters	HS Sections	Increase in input sourcing costs (in %)	Share of imports from non-TCA countries
		(1)	(2)
[1, 5]	Animals	0.4	16.5
[6, 14]	Vegetables	6.3	7.3
[15, 15]	Fats and Oils	2.7	22.4
[16, 24]	Prepared Foodstuffs	4.9	15.9
[25, 27]	Minerals	14.3	64.6
[28, 38]	Chemicals	8.7	13.1
[39, 40]	Plastics, Rubber and Articles	10.6	13.1
[41, 43]	Hides, Leather and Articles	11.1	9.1
[44, 46]	Wood, Straw and Articles	6.0	10.3
[47, 49]	Pulp, Paper and Articles	10.6	12.2
[50, 63]	Textiles and Articles	11.4	11.3
[64, 67]	Footwear, Headgear, Other	12.1	28.5
[68, 70]	Stone, Plaster, Cement, Other	10.9	6.9
[71, 71]	Precious Stones, Metals, Jewellery	8.7	37.0
[72, 83]	Base Metals and Articles	10.7	10.9
[84, 85]	Machinery, Electrical Equipment, Electronics	13.5	17.1
[86, 89]	Vehicles, Aircraft, Vessels	11.2	10.1
[90, 92]	Optical, Medical and Other Instruments	16.7	37.4
[94, 96]	Miscellaneous Manufactures	12.8	14.3
	Average	9.7	18.8

Note: Column (1) reports the average increase in expected input sourcing costs (in percentage) due to compliance with TCA RoO, across inputs within an HS section. Column 2 reports the corresponding average share of inputs imported from non-TCA countries in the pre-TCA period.

Figure 8: Firm-level expected increase in input sourcing costs and share of inputs imported from non-TCA countries



Notes: The figure reports the correlation between firms' expected increase in sourcing cost (binned) when complying with RoO, and the share of their imported inputs from non-TCA origins (binned).

8 Conclusions

Preferential RoO in FTAs restrict firms' sourcing decisions, preventing them from sourcing inputs from more efficient suppliers outside the FTA, in order to avoid losing origin status on their final goods.

To study firms' decisions to comply with RoO and the effects of these sourcing restrictions, we develop a three-country model of trade, in which heterogeneous firms produce differentiated varieties of a final good using labor and a composite intermediate input restricted by RoO.

In our empirical analysis, we leverage Brexit as a shock in trade relations between the EU and the UK (from the EU Customs Union to the TCA), which implies that since 2021 tariff free trade between them is conditional on compliance with TCA RoO. We construct a rich dataset on TCA RoO with administrative data from France to study the effects of these sourcing restrictions on French firms.

We first examine the effects of TCA RoO on firms' input sourcing decisions. Our model predicts that the introduction of these rules in 2021 should decrease the incentives of French firms to source inputs from outside the TCA, and this effect should increase with their compliance gains. Exploiting within-firm cross-product and cross-country variation in treatment over time, we show that the introduction of TCA RoO significantly reduced the probability that French firms source inputs from third countries. As expected, the effects increase with firms' compliance gains. Placebo tests confirm that RoO have no effect when firms have no incentives to comply with them. The event study specification shows that the negative effects of RoO become larger over time, suggesting that it takes time for firms to switch suppliers. Our baseline estimates imply that a one standard deviation increase in the number of RoO restrictions faced by a firm on its inputs decreases the average probability of importing from non-TCA countries by up to 29%.

Our model also predicts that, by distorting firms' sourcing decisions, RoO should have negative effects of firms' global revenues: firms that stop sourcing restricted inputs from third countries to comply with RoO should experience an increase in their marginal cost and a fall in the sales of their final good in all markets. In line with this prediction, we show that French firms that adjusted their input sourcing to comply with TCA RoO (constrained compliers) experienced a reduction in their global sales compared to firms that could obtain origin without changing their sourcing strategies (unconstrained compliers). We also show that, by distorting their supply chains, TCA RoO led to a reduction in firms' employment and productivity.

By combining the structure of our theoretical model with our data, we can infer the

expected cost of complying with TCA RoO. We find that compliance with these sourcing restrictions increased the expected input sourcing costs of French firms by 9.7% on average.

It would be interesting to study the effects of TCA RoO on UK firms. We expect these effects to be much larger, since UK firms have much stronger incentives to comply with the rules than French firms: obtaining origin on their final goods allows them to maintain duty-free access in 27 EU countries (compared to 1 for French firms). In the case of the UK, we also expect large negative effects on foreign direct investment. This is because multinational corporations like Nissan — which before Brexit used the UK as an export platform for the rest of the EU — have strong incentives to relocate to the EU Customs Union so as obtain unconditional duty free treatment for their final goods.

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A-1 Descriptive Statistics

Table A-1: Number of RoO on HS6 final goods with CTC and VA requirements

HS Chapters	HS Sections	<i>N</i> goods	CTC	Additional VA req.	Alternative VA req.	Only VA req.
[1, 5]	Animals	371	356	0	0	0
[6, 14]	Vegetables	308	253	0	0	0
[15, 15]	Fats and Oils	48	45	0	0	0
[16, 24]	Prepared Foodstuffs	214	211	4	0	0
[25, 27]	Minerals	147	110	0	68	0
[28, 38]	Chemicals	868	861	26	846	0
[39, 40]	Plastics, Rubber and Articles	209	209	0	205	0
[41, 43]	Hides, Leather and Articles	69	69	0	31	0
[44, 46]	Wood, Straw and Articles	121	121	0	121	0
[47, 49]	Pulp, Paper and Articles	141	141	0	141	0
[50, 63]	Textiles and Articles	803	799	6	0	4
[64, 67]	Footwear, Headgear, Other	47	47	0	6	0
[68, 70]	Stone, Plaster, Cement, Other	143	143	0	100	0
[71, 71]	Precious Stones, Metals, Jewellery	53	34	0	0	0
[72, 83]	Base Metals and Articles	563	505	41	99	0
[84, 85]	Machinery, Electrical Equipment, Electronics	781	750	0	750	31
[86, 89]	Vehicles, Aircraft, Vessels	143	100	0	100	43
[90, 92]	Optical, Medical and Other Instruments	210	193	0	193	17
[93, 93]	Arms and Ammunition	18	0	0	0	18
[94, 96]	Miscellaneous Manufactures	122	120	0	120	2
[97, 97]	Works of Art	7	7	0	0	0
All		5,386	5,074	77	2,780	115

Note: Number of final goods at the HS 6-digit level by HS Section, identified using groups of HS chapters (HS-2). The column “CTC” shows the number of final goods with CTC requirements. “Additional VA req.” shows the number of final goods that have VA requirements in the RoO on top of CTC requirements. “Alternative VA req.” shows the number of final goods that have VA requirements as alternative requirements to CTC. The column “Only VA req.” shows the number of final goods that have only VA requirements in the RoO.

Table A-2: Descriptive statistics of RoO_j

HS Chapters	HS Sections	N	Mean	Standard Deviation	Min.	Max.
[1, 5]	Animals	356	201.26	83.77	75.00	265.00
[6, 14]	Vegetables	269	46.99	29.92	1.00	101.00
[15, 15]	Fats and Oils	48	4.25	0.84	3.00	6.00
[16, 24]	Prepared Foodstuffs	200	15.15	15.55	1.00	42.00
[25, 27]	Minerals	110	3.31	2.26	1.00	9.00
[28, 38]	Chemicals	862	1.27	1.15	1.00	9.00
[39, 40]	Plastics, Rubber and Articles	209	4.12	4.04	1.00	14.00
[41, 43]	Hides, Leather and Articles	69	4.97	3.52	1.00	12.00
[44, 46]	Wood, Straw and Articles	121	9.02	6.37	1.00	18.00
[47, 49]	Pulp, Paper and Articles	141	6.12	3.71	1.00	12.00
[50, 63]	Textiles and Articles	803	679.38	176.08	176.00	759.00
[64, 67]	Footwear, Headgear, Other	25	5.00	7.62	1.00	25.00
[68, 70]	Stone, Plaster, Cement, Other	143	4.76	3.17	1.00	13.00
[71, 71]	Precious Stones, Metals, Jewellery	34	8.24	6.13	2.00	15.00
[72, 83]	Base Metals and Articles	506	30.20	31.07	1.00	109.00
[84, 85]	Machinery, Electrical Equipment, Electronics	750	8.61	8.95	1.00	73.00
[86, 89]	Vehicles, Aircraft, Vessels	100	10.02	6.88	1.00	23.00
[90, 92]	Optical, Medical and Other Instruments	193	5.87	3.14	1.00	13.00
[93, 93]	Arms and Ammunition	-	-	-	-	-
[94, 96]	Miscellaneous Manufactures	121	7.74	5.24	1.00	17.00
[97, 97]	Works of Art	7	1.29	0.49	1.00	2.00
All		5,067	130.96	254.28	1.00	759.00

Note: Descriptive statistics for RoO_j by HS Section, identified as groups of HS chapters (HS-2). The variable RoO_j is defined in equation (19).

Table A-3: Descriptive statistics of RoO_i

HS Chapters	HS Sections	N	Mean	Standard Deviation	Min.	Max.
[1, 5]	Animals	356	165.61	78.86	32.00	224.00
[6, 14]	Vegetables	253	48.28	26.59	1.00	69.00
[15, 15]	Fats and Oils	45	10.13	32.95	1.00	132.00
[16, 24]	Prepared Foodstuffs	211	73.96	121.63	1.00	366.00
[25, 27]	Minerals	110	3.40	2.36	1.00	9.00
[28, 38]	Chemicals	861	1.60	3.77	1.00	36.00
[39, 40]	Plastics, Rubber and Articles	209	4.12	4.08	1.00	14.00
[41, 43]	Hides, Leather and Articles	69	4.97	3.65	1.00	12.00
[44, 46]	Wood, Straw and Articles	121	9.02	6.37	1.00	18.00
[47, 49]	Pulp, Paper and Articles	141	6.12	3.71	1.00	12.00
[50, 63]	Textiles and Articles	799	682.78	172.37	1.00	804.00
[64, 67]	Footwear, Headgear, Other	47	2.66	0.87	1.00	4.00
[68, 70]	Stone, Plaster, Cement, Other	143	4.76	3.39	1.00	13.00
[71, 71]	Precious Stones, Metals, Jewellery	34	8.24	6.13	2.00	15.00
[72, 83]	Base Metals and Articles	505	30.26	49.34	1.00	198.00
[84, 85]	Machinery, Electrical Equipment, Electronics	750	8.61	4.70	1.00	19.00
[86, 89]	Vehicles, Aircraft, Vessels	100	10.02	5.10	1.00	18.00
[90, 92]	Optical, Medical and Other Instruments	193	5.87	3.14	1.00	13.00
[93, 93]	Arms and Ammunition	-	-	-	-	-
[94, 96]	Miscellaneous Manufactures	120	7.81	5.26	1.00	17.00
[97, 97]	Works of Art	7	1.29	0.49	1.00	2.00
All		5,074	130.78	254.40	1.00	804.00

Note: Descriptive statistics for RoO_i by HS Section, identified as groups of HS chapters (HS-2). The variable RoO_i is defined in equation (20).

Table A-4: Descriptive statistics of *Applied Tariff_{jot}*

HS Chapters	HS Sections	N	Mean	Std.	Min.	Max.
[1, 5]	Animals	667,800	5.13	12.89	0.00	186.48
[6, 14]	Vegetables	554,400	3.51	11.93	0.00	350.64
[15, 15]	Fats and Oils	86,400	2.96	9.20	0.00	225.88
[16, 24]	Prepared Foodstuffs	385,200	7.14	17.24	0.00	227.06
[25, 27]	Minerals	266,400	0.08	0.39	0.00	6.07
[28, 38]	Chemicals	1,562,400	1.22	3.09	0.00	83.23
[39, 40]	Plastics, Rubber and Articles	376,200	1.16	2.32	0.00	6.50
[41, 43]	Hides, Leather and Articles	124,200	0.70	1.63	0.00	9.00
[44, 46]	Wood, Straw and Articles	217,800	0.48	1.56	0.00	10.00
[47, 49]	Pulp, Paper and Articles	253,800	0.00	0.00	0.00	0.00
[50, 63]	Textiles and Articles	1,445,400	2.15	3.82	0.00	12.00
[93, 93]	Arms and Ammunition	32,400	1.41	1.43	0.00	3.20
[64, 67]	Footwear, Headgear, Other	84,600	1.94	4.33	0.00	17.00
[68, 70]	Stone, Plaster, Cement, Other	257,400	0.93	2.18	0.00	12.00
[71, 71]	Precious Stones, Metals, Jewellery	95,400	0.14	0.66	0.00	4.00
[72, 83]	Base Metals and Articles	1,013,400	0.51	1.42	0.00	9.00
[84, 85]	Machinery, Electrical Equipment, Electronics	1,405,800	0.43	1.13	0.00	14.00
[86, 89]	Vehicles, Aircraft, Vessels	257,400	1.14	2.80	0.00	17.37
[90, 92]	Optical, Medical and Other Instruments	378,000	0.50	1.26	0.00	6.70
[93, 93]	Arms and Ammunition	-	-	-	-	-
[94, 96]	Miscellaneous Manufactures	219,600	0.65	1.41	0.00	7.70
[97, 97]	Works of Art	12,600	0.00	0.00	0.00	0.00
	All	9,696,600	1.68	6.34	0.00	350.64

Note: Applied tariff rates by France on imports of good j from country o at time t .

Table A-5: Descriptive statistics of τ_i^{UK}

HS	industry	N	Mean	Std.	Min.	Max.
[1, 5]	Animals	370	13.23	13.46	0.00	121.12
[6, 14]	Vegetables	308	8.67	14.76	0.00	170.72
[15, 15]	Fats and Oils	48	6.81	8.43	0.00	46.00
[16, 24]	Prepared Foodstuffs	214	18.20	19.72	0.00	125.79
[25, 27]	Minerals	147	0.18	0.49	0.00	3.62
[28, 38]	Chemicals	866	3.73	3.28	0.00	58.23
[39, 40]	Plastics, Rubber and Articles	209	4.22	2.29	0.00	6.37
[41, 43]	Hides, Leather and Articles	69	2.00	1.80	0.00	8.50
[44, 46]	Wood, Straw and Articles	121	1.48	2.46	0.00	10.00
[47, 49]	Pulp, Paper and Articles	140	0.00	0.00	0.00	0.00
[50, 63]	Textiles and Articles	794	7.17	3.85	0.00	12.00
[64, 67]	Footwear, Headgear, Other	47	6.78	5.94	0.00	16.50
[68, 70]	Stone, Plaster, Cement, Other	139	2.73	3.07	0.00	12.00
[71, 71]	Precious Stones, Metals, Jewellery	53	0.57	1.21	0.00	4.00
[72, 83]	Base Metals and Articles	561	1.32	1.73	0.00	8.25
[84, 85]	Machinery, Electrical Equipment, Electronics	777	1.18	1.43	0.00	14.00
[86, 89]	Vehicles, Aircraft, Vessels	143	3.65	3.94	0.00	14.50
[90, 92]	Optical, Medical and Other Instruments	207	1.26	1.34	0.00	6.00
[93, 93]	Arms and Ammunition	18	1.76	1.06	0.00	2.60
[94, 96]	Miscellaneous Manufactures	122	2.01	1.23	0.00	4.80
[97, 97]	Works of Art	7	0.00	0.00	0.00	0.00
All		5,360	4.74	8.18	0.00	170.72

Note: This table reports descriptive statistics for MFN tariff rates applied by the UK in 2022.

Table A-6: In-sample descriptive statistics

	Obs.	Mean	Std. dev.	Min	Max
Full sample (all suppliers)					
RoO_{fj}	4,983,209	6.65	21.85	0	219
RoO_{fj} (ln)	4,983,209	0.84	1.17	0	5.39
$Compliance\ Gains_{fj}$ (ln)	4,983,209	0.005	0.013	0	0.58
$Applied\ Tariff_{jot}$ (ln)	4,983,209	0.006	0.016	0	0.86
Firms w/o VA rules (all suppliers)					
RoO_{fj}	4,431,340	6.84	22.43	0	219
RoO_{fj} (ln)	4,431,340	0.84	1.17	0	5.39
$Compliance\ Gains_{fj}$ (ln)	4,431,340	0.005	0.013	0	0.58
$Applied\ Tariff_{jot}$ (ln)	4,431,340	0.006	0.016	0	0.86
Full sample (actual suppliers)					
RoO_{fj}	1,501,847	5.04	18.25	0	219
RoO_{fj} (ln)	1,501,847	0.79	1.05	0	5.36
$Compliance\ Gains_{fj}$ (ln)	1,501,847	0.004	0.011	0	0.53
$Applied\ Tariff_{jot}$ (ln)	1,501,847	0.006	0.015	0	0.52
Firms w/o VA rules (actual suppliers)					
RoO_{fj}	1,322,951	5.18	18.79	0	219
RoO_{fj} (ln)	1,322,951	0.79	1.06	0	5.39
$Compliance\ Gains_{fj}$ (ln)	1,322,951	0.004	0.011	0	0.53
$Applied\ Tariff_{jot}$ (ln)	1,322,951	0.006	0.016	0	0.52

Note: RoO variable rescaled by the minimum RoO number in the sample. Descriptive statistics in panel a (panel b) based on the set of firms in column (1) (column (2)) of Table 1.

A-2 Additional results and robustness checks

Table A-7: Effects of TCA RoO on firms' imports of intermediate inputs
(adding product-country-year fixed effects)

Dep var:	$Pr(Imports_{fjot})$	
	(1)	(2)
	All CTC rules	No VA rules
$Non-TCA_{ot} \times RoO_{fj} \times Compliance\ Gains_{fj}$	-1.411 ⁺	-2.148**
	(0.934)	(0.956)
$Non-TCA_{ot} \times Compliance\ Gains_{fj}$	-4.133***	-4.888***
	(1.169)	(1.196)
$Non-TCA_{ot} \times RoO_{fj}$	0.002	0.008
	(0.015)	(0.016)
Estimator	PPML	PPML
Firm-Country-Year FE	Yes	Yes
Firm-Product-Year FE	Yes	Yes
Country-Product-Year FE	Yes	Yes
Observations	4,841,679	4,276,108

Notes: The dependent variable is an indicator equal to 1 if firm f imports intermediate input j from origin country o in year t , and 0 otherwise. RoO_{fj} is the number of firm f 's final goods imposing CTC RoO on input j (in log). $Non-TCA_{ot}$ is an indicator identifying non-TCA countries in the post-TCA period (i.e., from 2021). $Compliance\ Gains_{fj}$, defined in equation (24), is included in logarithmic form as $\log(1 + Compliance\ Gains_{fj})$. $Applied\ Tariff_{jot}$ is the tariff applied by France on imports of good j from origin country o in year t and enters the specification as $\log(1 + Applied\ Tariff_{jot})$. The sample includes all origin countries o that accounted for at least 1% of pre-TCA imports of input j ; in column (1), it includes all firms that face the CTC RoO; in column (2), it excludes firms that are constrained by both CTC and VA sourcing requirements. Standard errors adjusted for clustering by firm in all specifications. ***, **, *, + significantly different from 0 at the 1%, 5%, 10%, and 15% levels respectively.

Table A-8: Effects of RoO on firms' imports
(vertically-related goods only)

Dep var:	$Pr(Imports_{fjot})$	
	(1)	(2)
	All CTC rules	No VA rules
$Non-TCA_{ot} \times Compliance\ Gains_{fj}$	-3.062*** (1.047)	- 4.134*** (1.098)
$Non-TCA_{ot} \times RoO_{fj} \times Compliance\ Gains_{fj}$	-1.592** (0.795)	-1.942** (0.757)
$Non-TCA_{ot} \times RoO_{fj}$	0.009 (0.012)	0.010 (0.013)
$Applied\ Tariff_{jot}$	-0.754** (0.372)	-0.697* (0.410)
Estimator	PPML	PPML
Firm-Country-Year FE	Yes	Yes
Firm-Product-Year FE	Yes	Yes
Observations	4,883,598	4,340,621

Notes: The dependent variable is an indicator equal to 1 if firm f imports intermediate input j from origin country o in year t , and 0 otherwise. RoO_{fj} is the number of firm f 's final goods imposing CTC RoO on input j (in log). $Non-TCA_{ot}$ is an indicator identifying non-TCA countries in the post-TCA period (i.e., from 2021). $Compliance\ Gains_{fj}$, defined in equation (24), is included in logarithmic form as $\log(1 + Compliance\ Gains_{fj})$. $Applied\ Tariff_{jot}$ is the tariff applied by France on imports of good j from origin country o in year t and enters the specification as $\log(1 + Applied\ Tariff_{jot})$. The sample includes all origin countries o that accounted for at least 1% of pre-TCA imports of input j ; in column (1), it includes all firms that face the CTC RoO; in column (2), it excludes firms that are constrained by both CTC and VA sourcing requirements. Standard errors adjusted for clustering by firm in all specifications. ***, **, * significantly different from 0 at the 1%, 5%, and 10% levels respectively.

Table A-9: Effects of RoO on firms' imports
(excluding the UK from the origin countries)

Dep var:	$Pr(Imports_{fjot})$	
	(1)	(2)
	All CTC rules	No VA rules
$Non-TCA_{ot} \times Compliance\ Gains_{fj}$	-3.353*** (1.191)	- 4.514*** (1.250)
$Non-TCA_{ot} \times RoO_{fj} \times Compliance\ Gains_{fj}$	-1.776** (0.860)	-2.162** (0.863)
$Non-TCA_{ot} \times RoO_{fj}$	0.015 (0.013)	0.018 (0.015)
$Applied\ Tariff_{jot}$	-1.069*** (0.396)	-1.031** (0.436)
Estimator	PPML	PPML
Firm-Country-Year FE	Yes	Yes
Firm-Product-Year FE	Yes	Yes
Observations	4,364,332	3,881,214

Notes: The dependent variable is an indicator equal to 1 if firm f imports intermediate input j from origin country o in year t , and 0 otherwise. RoO_{fj} is the number of firm f 's final goods imposing CTC RoO on input j (in log). $Non-TCA_{ot}$ is an indicator identifying non-TCA countries in the post-TCA period (i.e., from 2021). $Compliance\ Gains_{fj}$, defined in equation (24), is included in logarithmic form as $\log(1 + Compliance\ Gains_{fj})$. $Applied\ Tariff_{jot}$ is the tariff applied by France on imports of good j from origin country o in year t and enters the specification as $\log(1 + Applied\ Tariff_{jot})$. The sample includes all origin countries o that accounted for at least 1% of pre-TCA imports of input j , excluding the UK; in column (1), it includes all firms that face the CTC RoO; in column (2), it excludes firms that are constrained by both CTC and VA sourcing requirements. Standard errors adjusted for clustering by firm in all specifications. ***, **, * significantly different from 0 at the 1%, 5%, and 10% levels respectively.